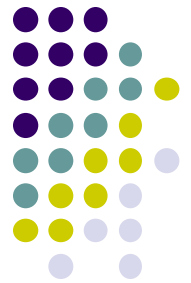


Q

Corneal Optics

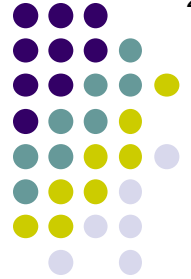


1

- What is the refractive index (n) of the cornea?

A

Corneal Optics



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This is a trickier question than it seems. The answer usually given is 1.34 (or 1.3375 by the OCD among us). But this is actually incorrect!

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Corneal Optics

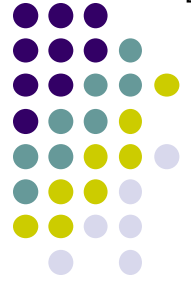


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To capture the refractive power of the cornea with a single value of n is a matter of considerable clinical utility. However, the task of assigning a single n to describe the cornea's refracting power is complicated by the effect of the cornea-aqueous interface on transmission of incoming light.

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A

Corneal Optics



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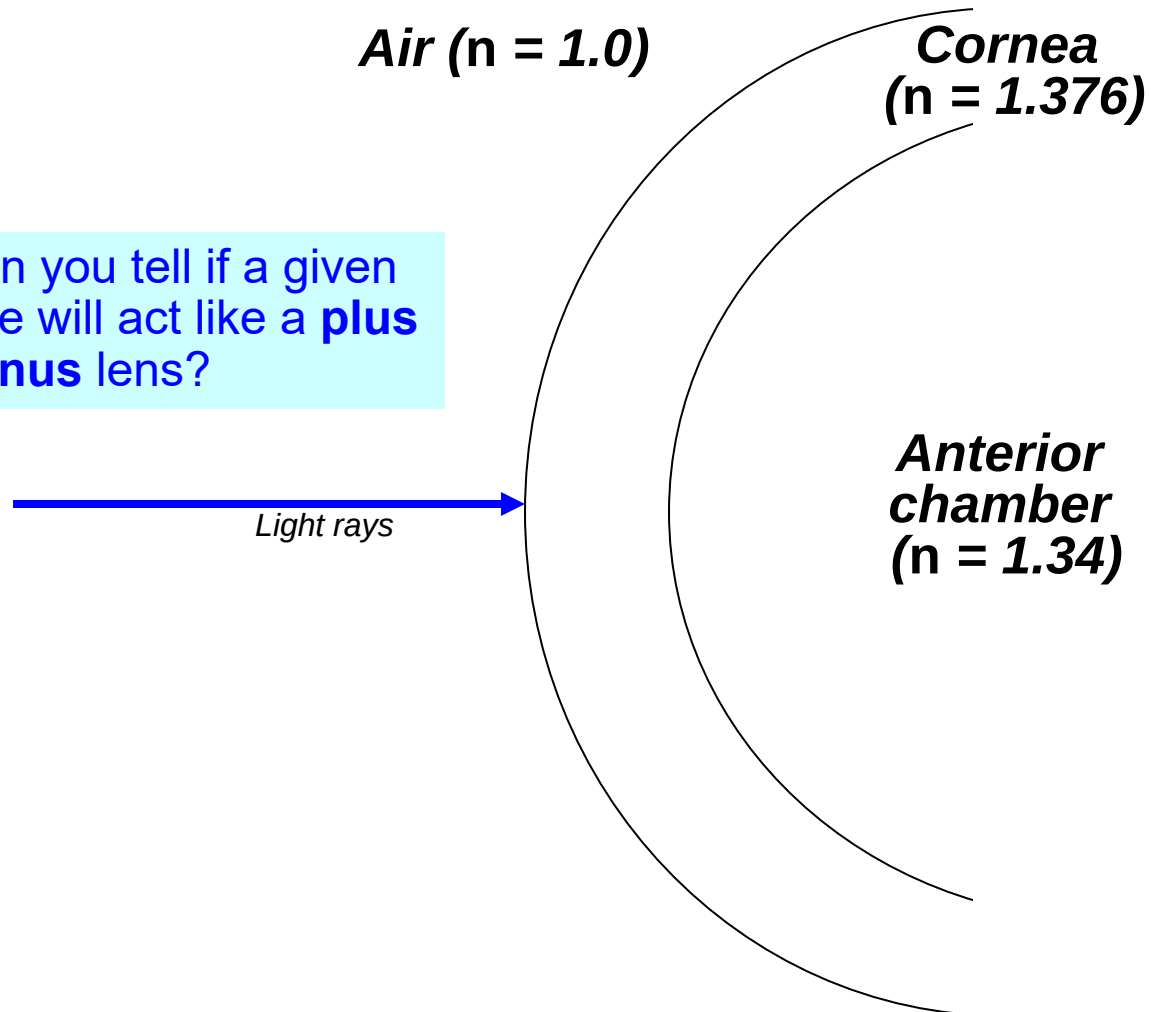
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The math is demonstrated on the following slides.

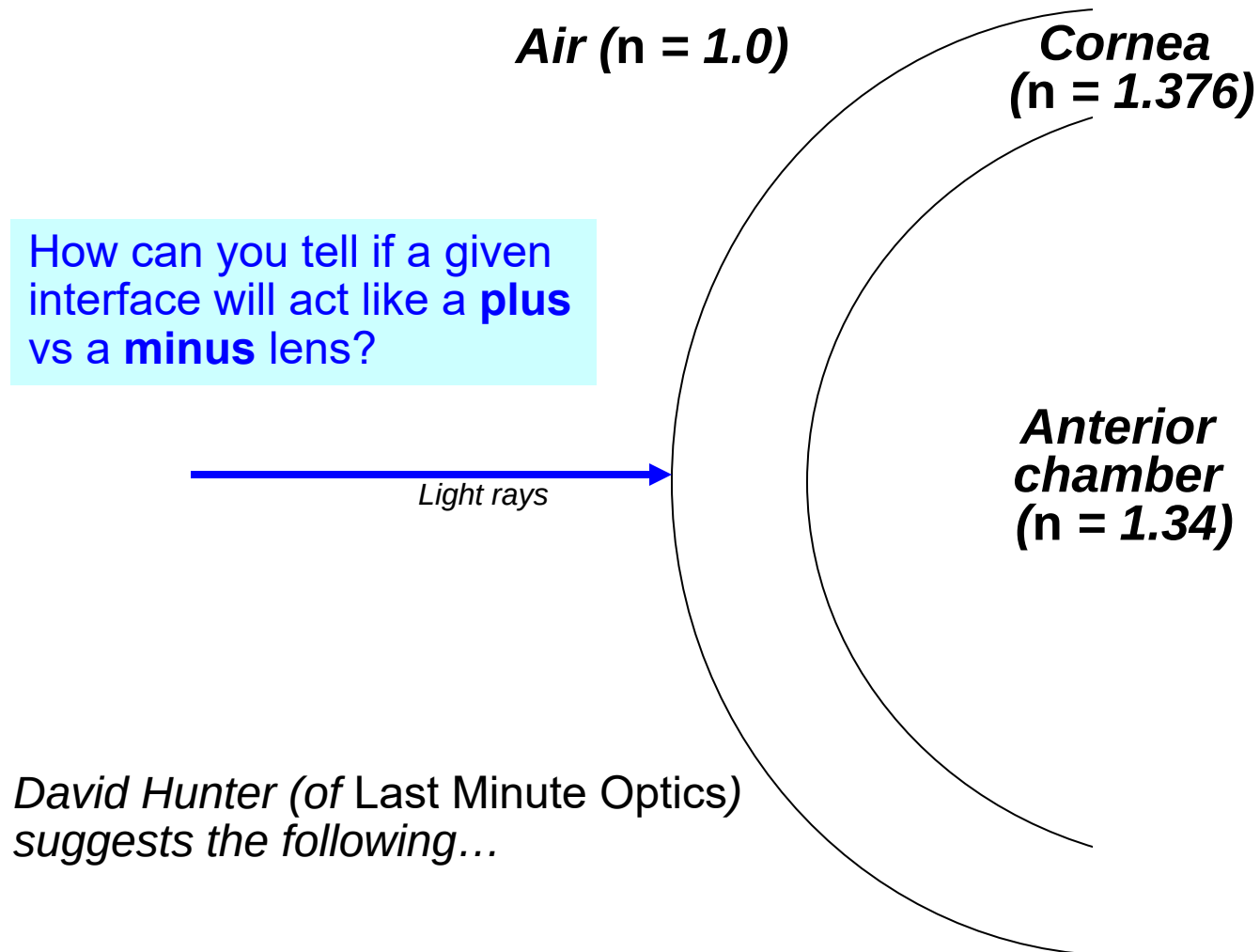
Corneal Optics



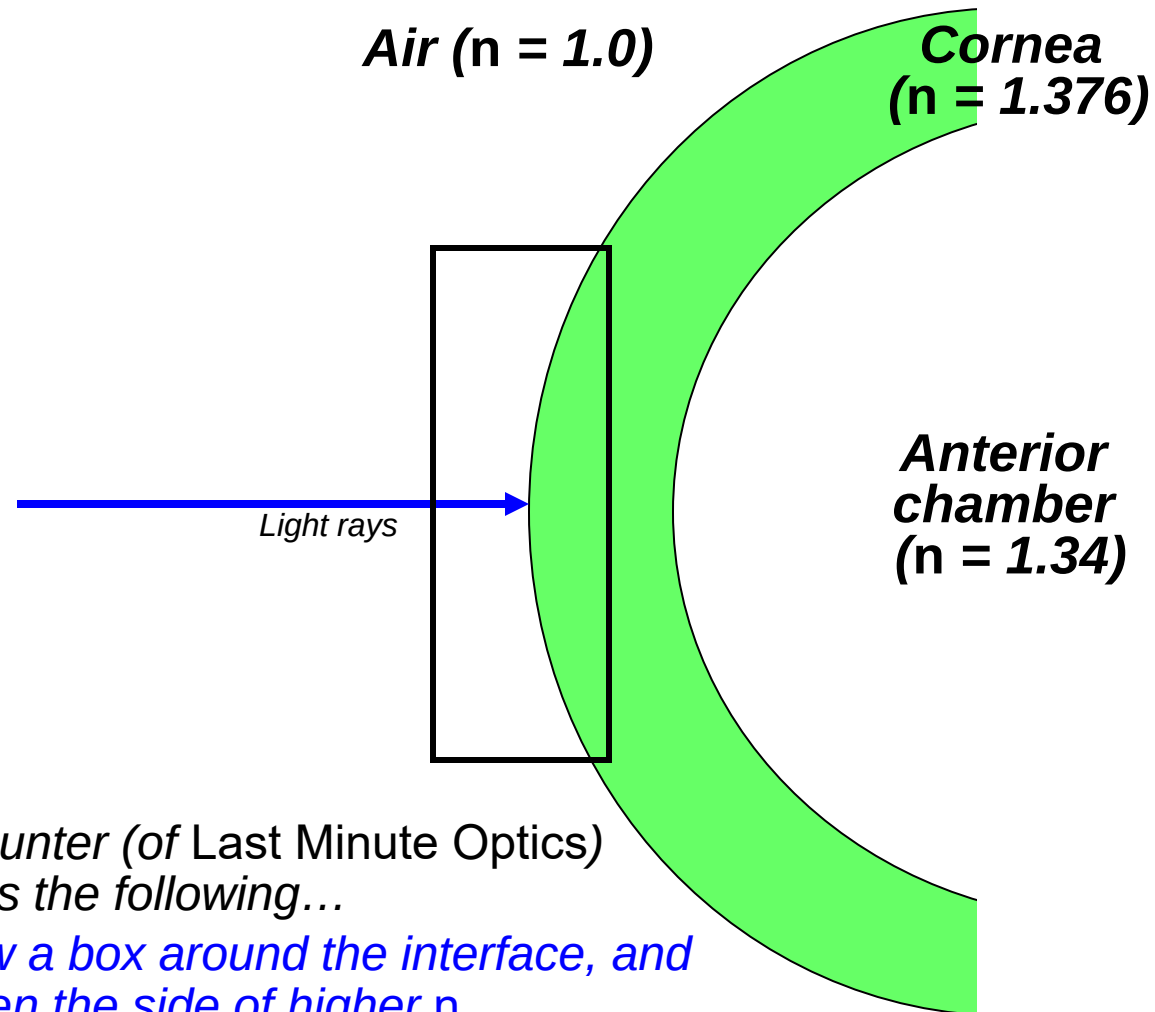
How can you tell if a given interface will act like a **plus** vs a **minus** lens?



Corneal Optics



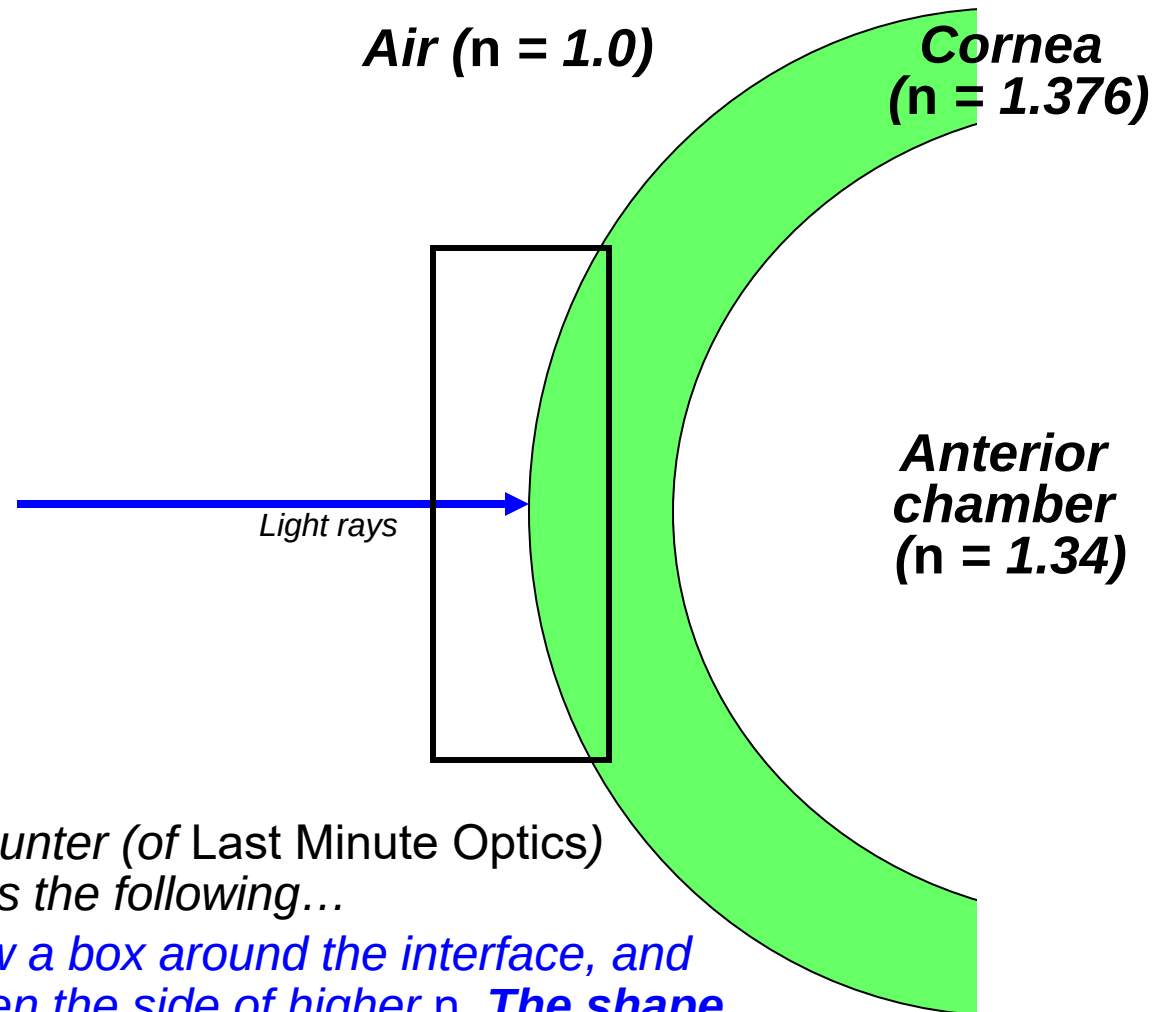
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David Hunter (of Last Minute Optics) suggests the following...

'Draw a box around the interface, and darken the side of higher n.'

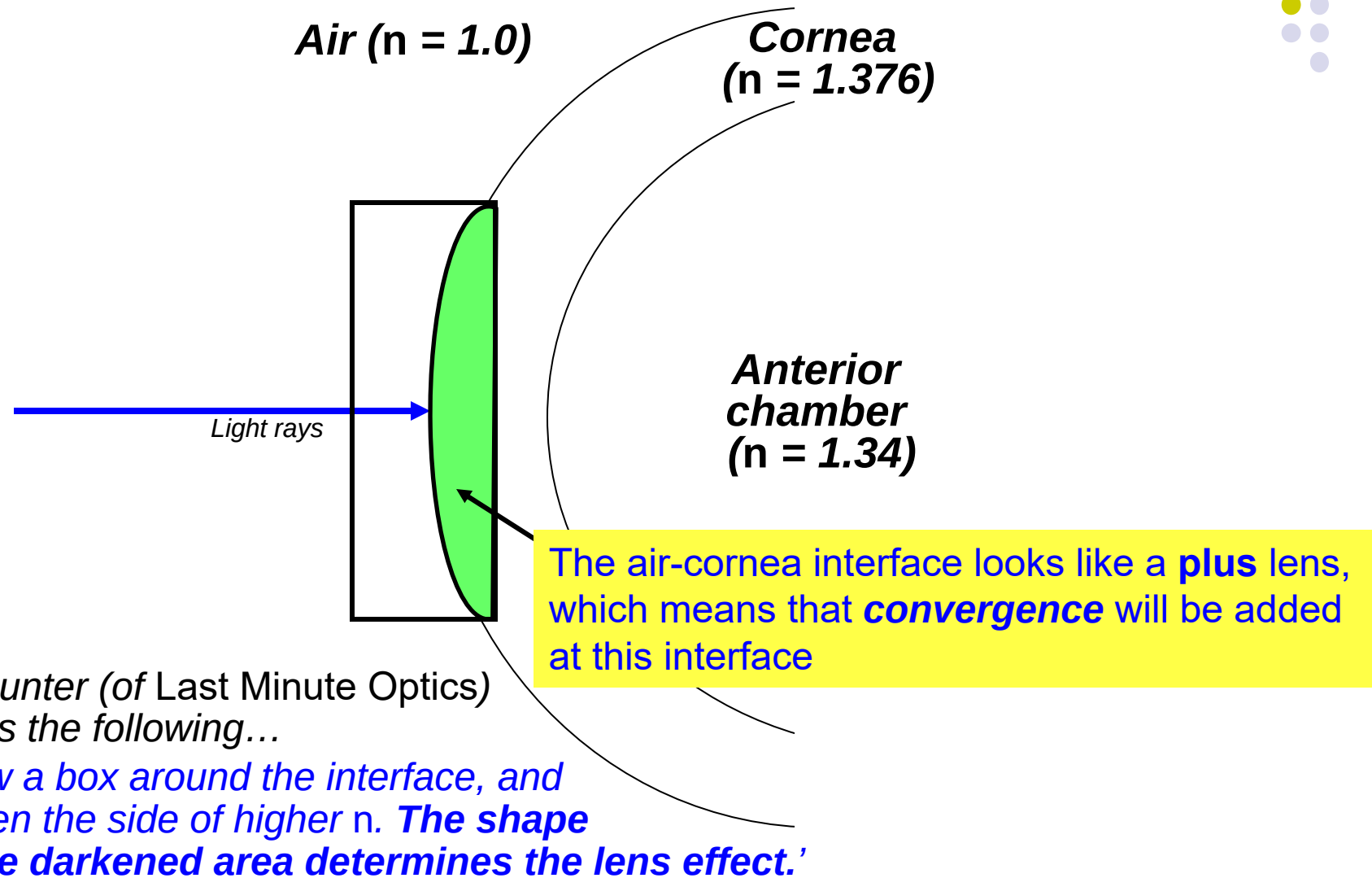
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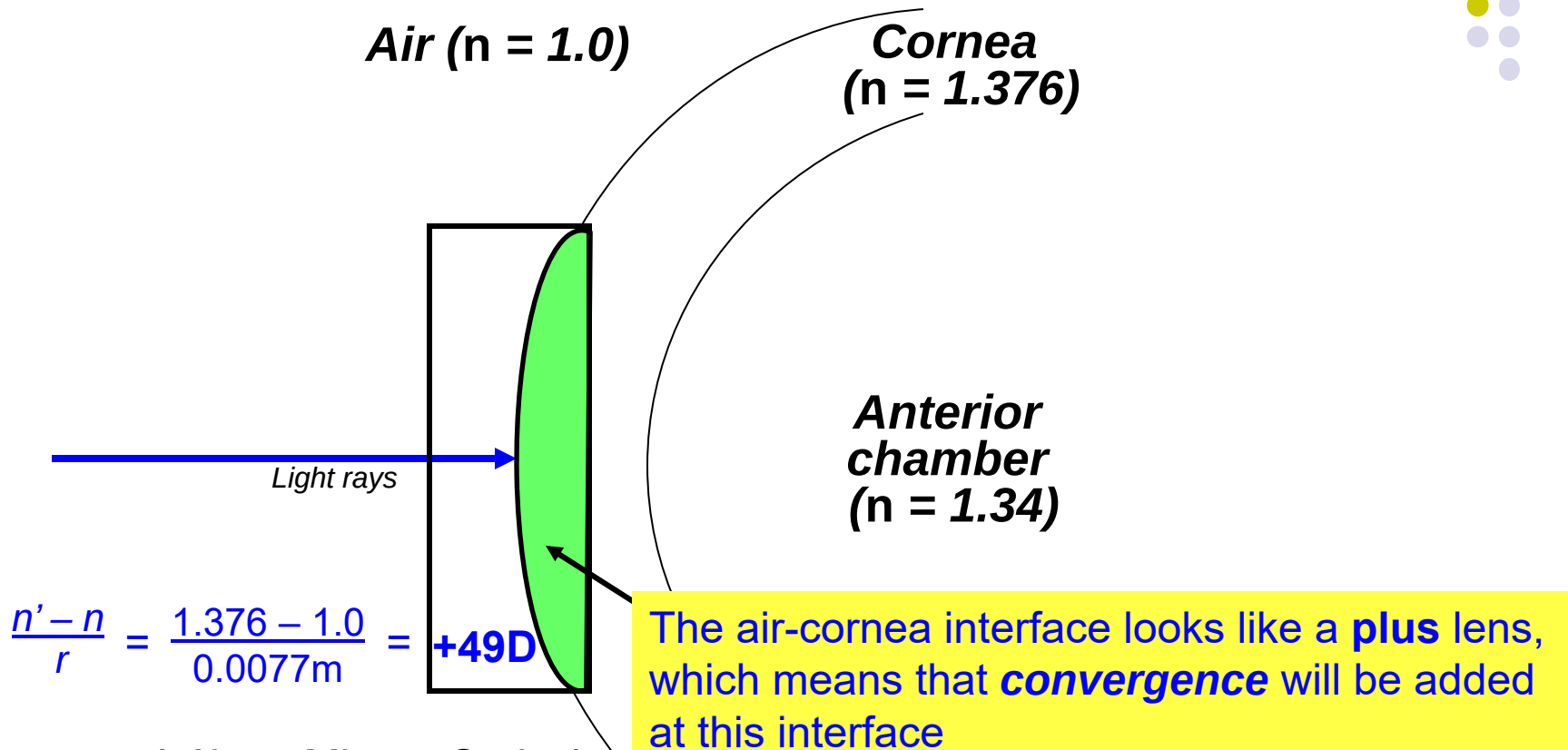
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Corneal Optics



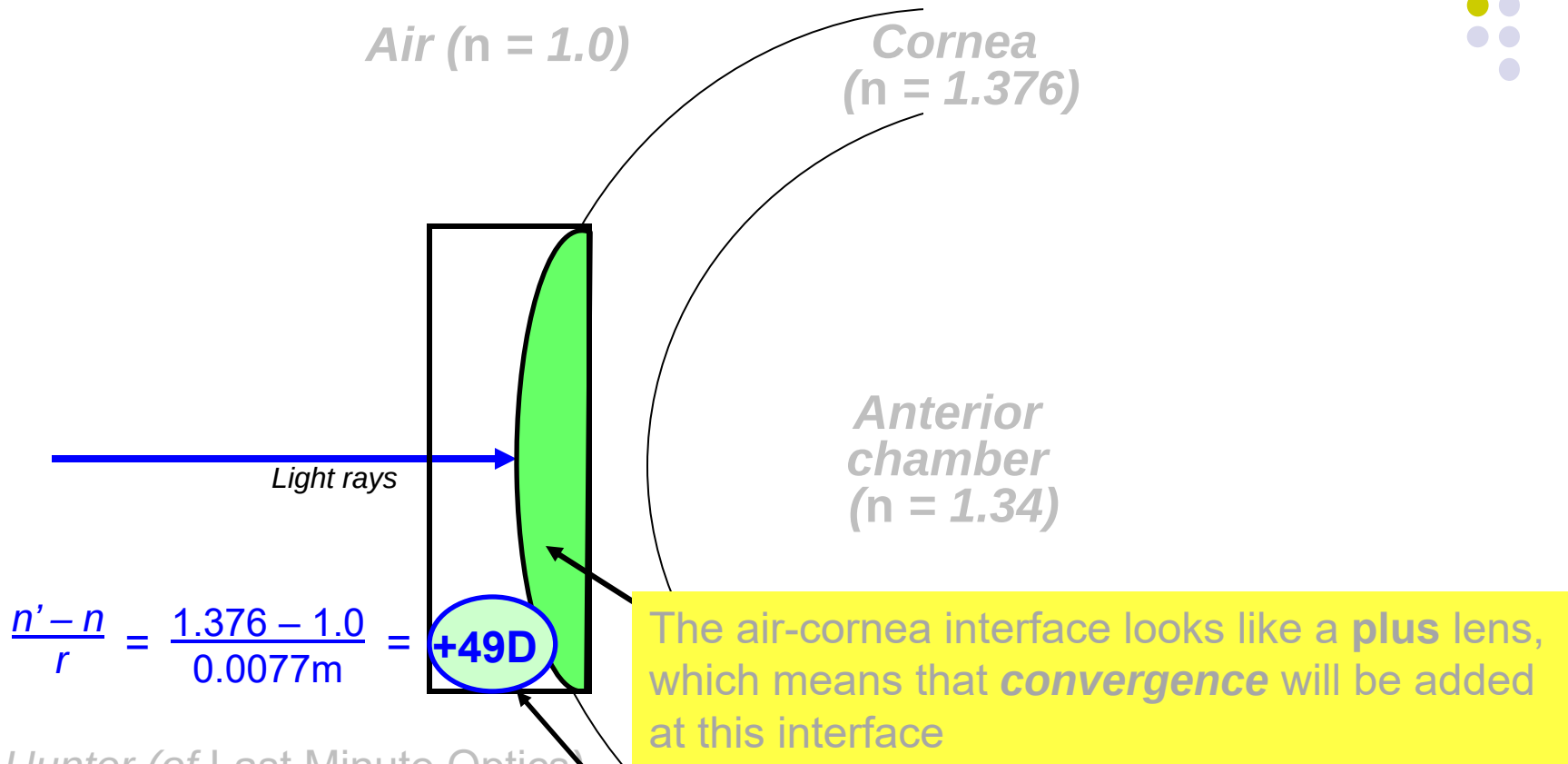
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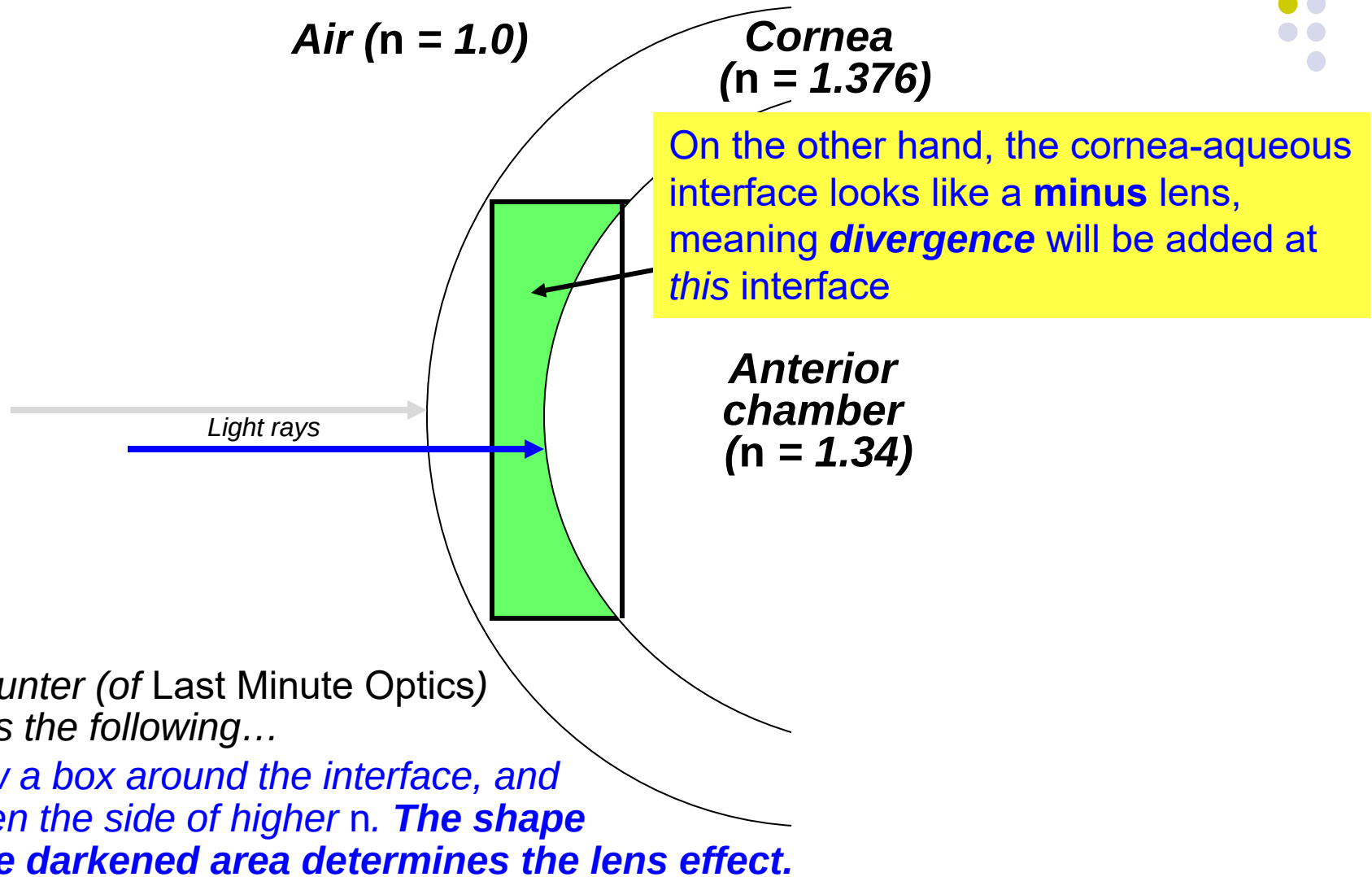


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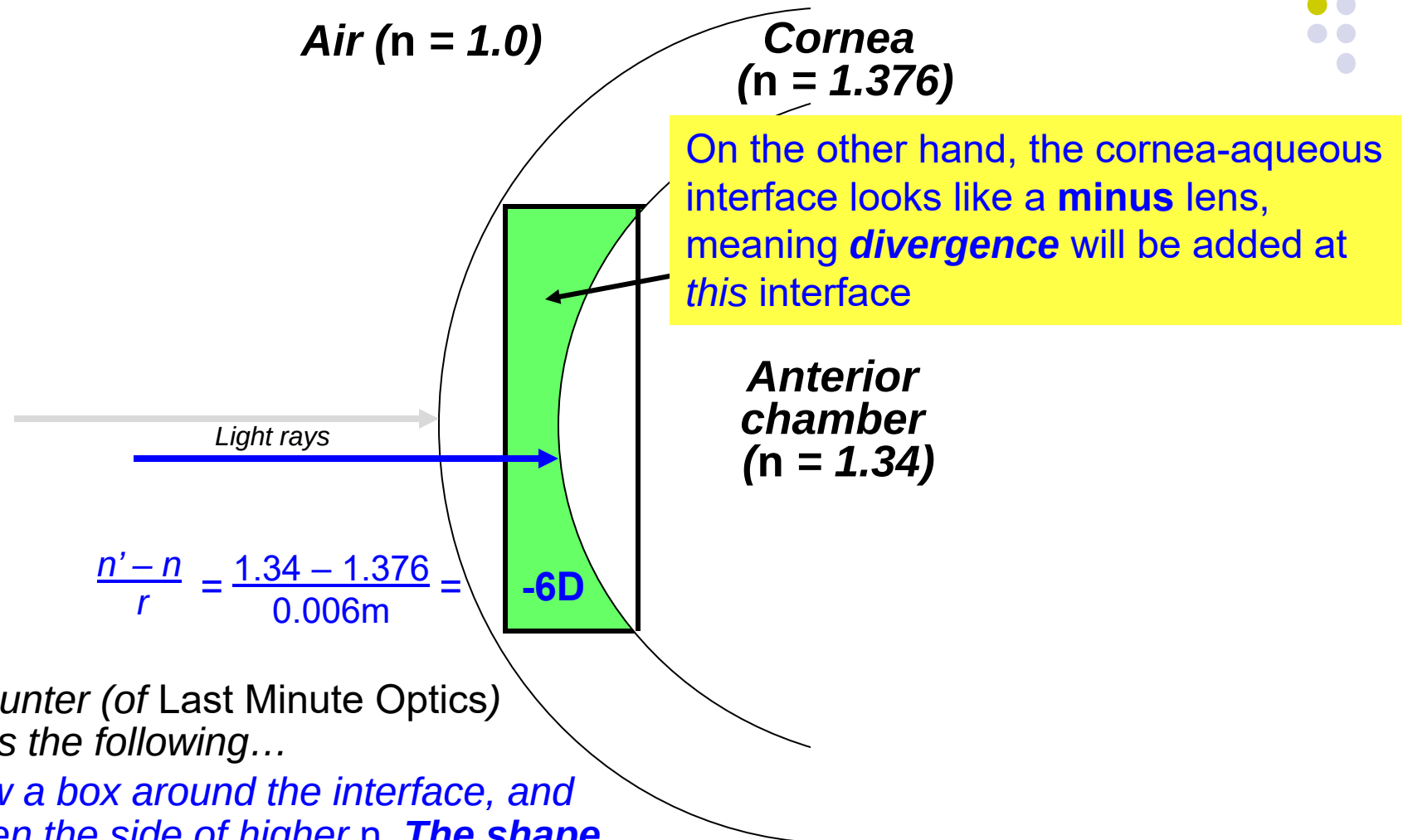
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Specifically, the effect is that of a plus lens to the tune of about +49D

Corneal Optics



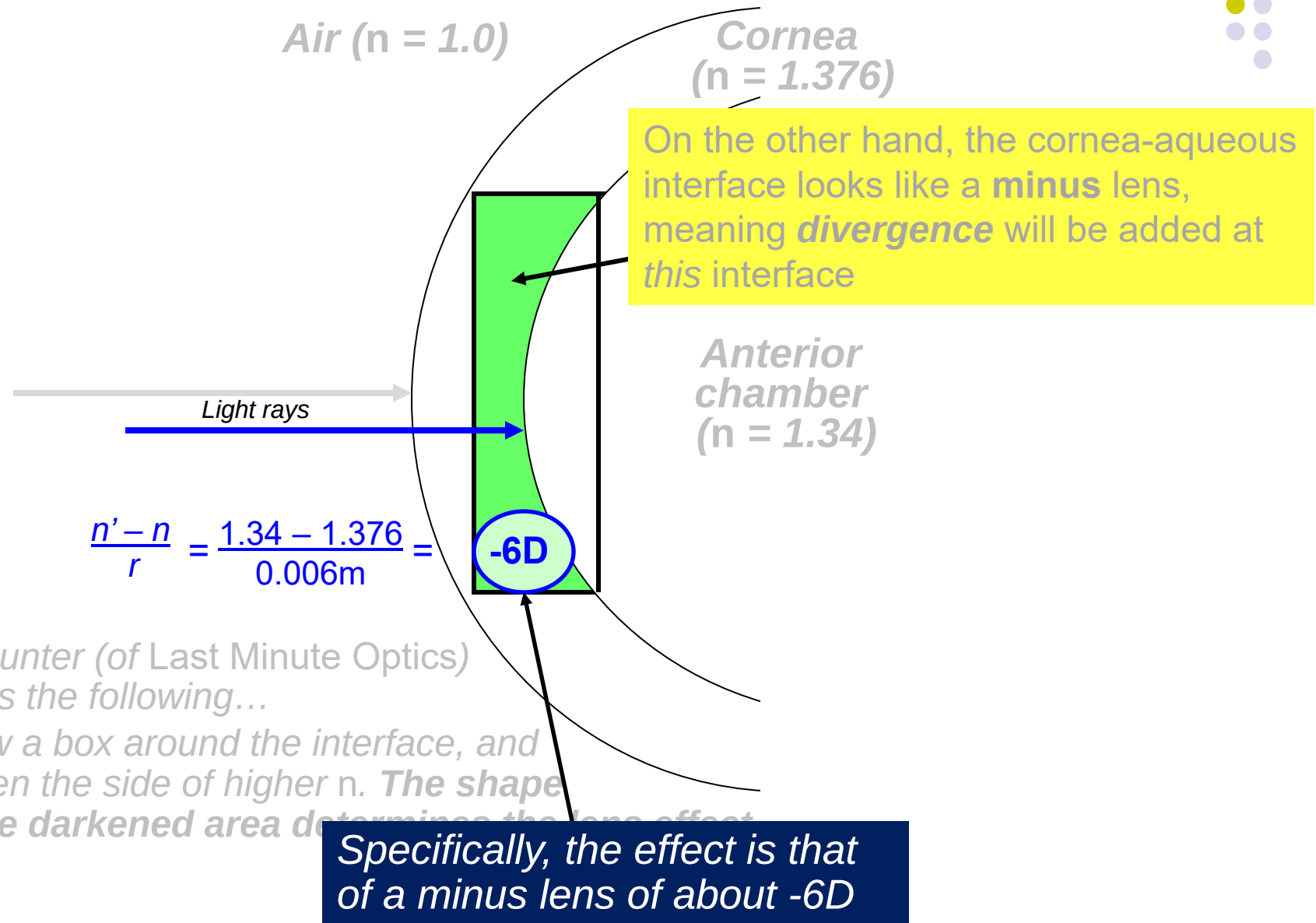
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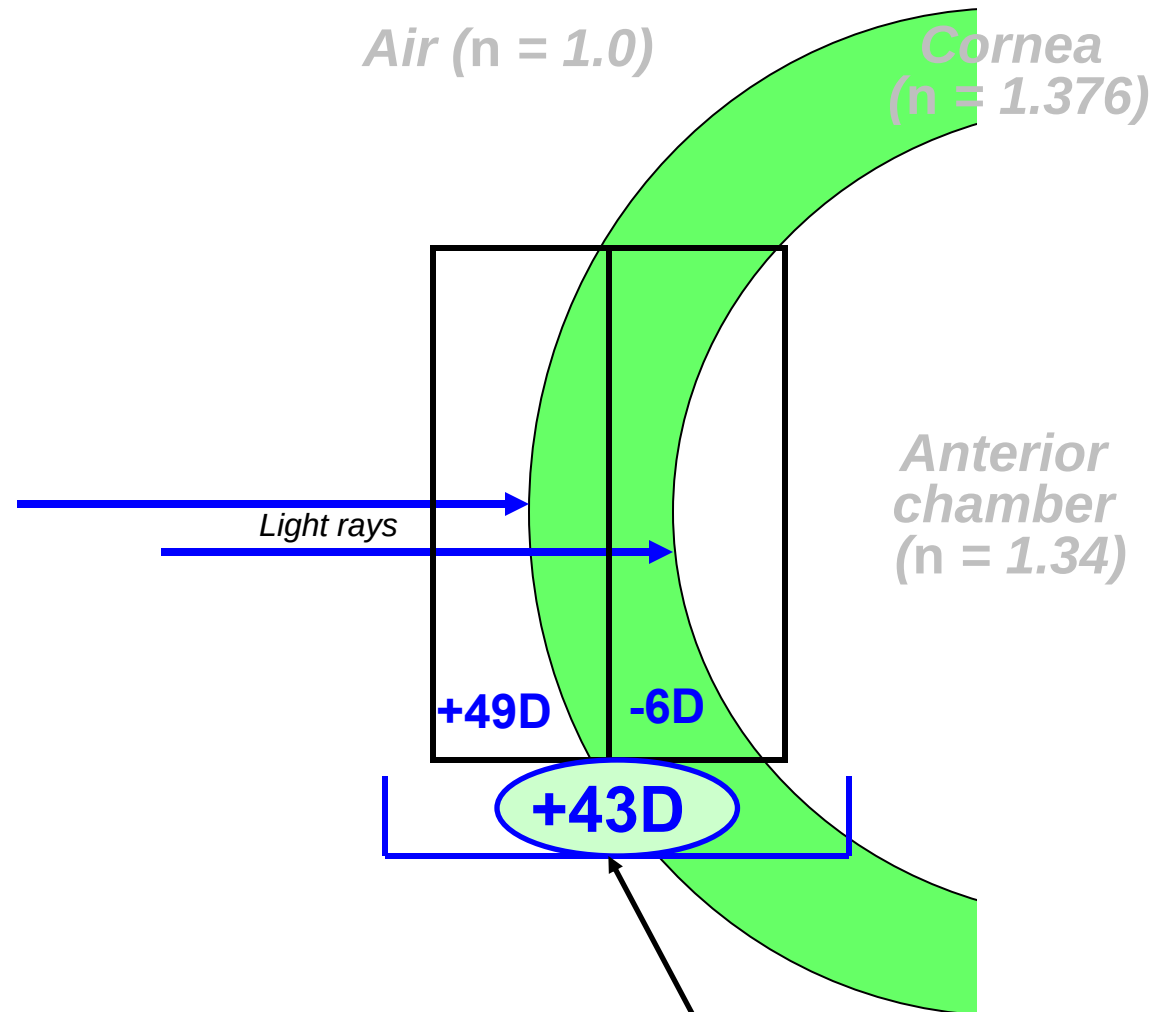
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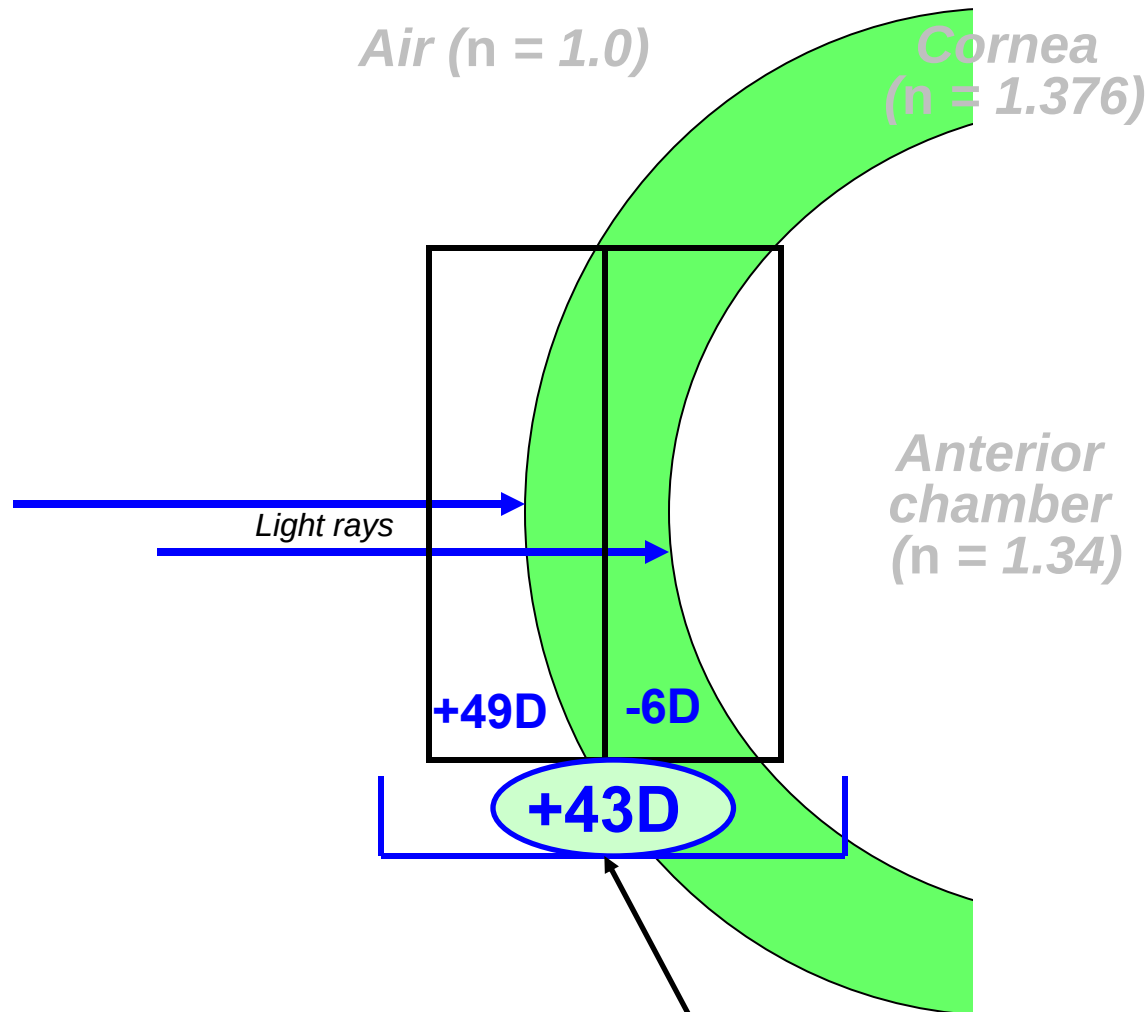


Corneal Optics



The net result across the cornea is an overall power of about 43D

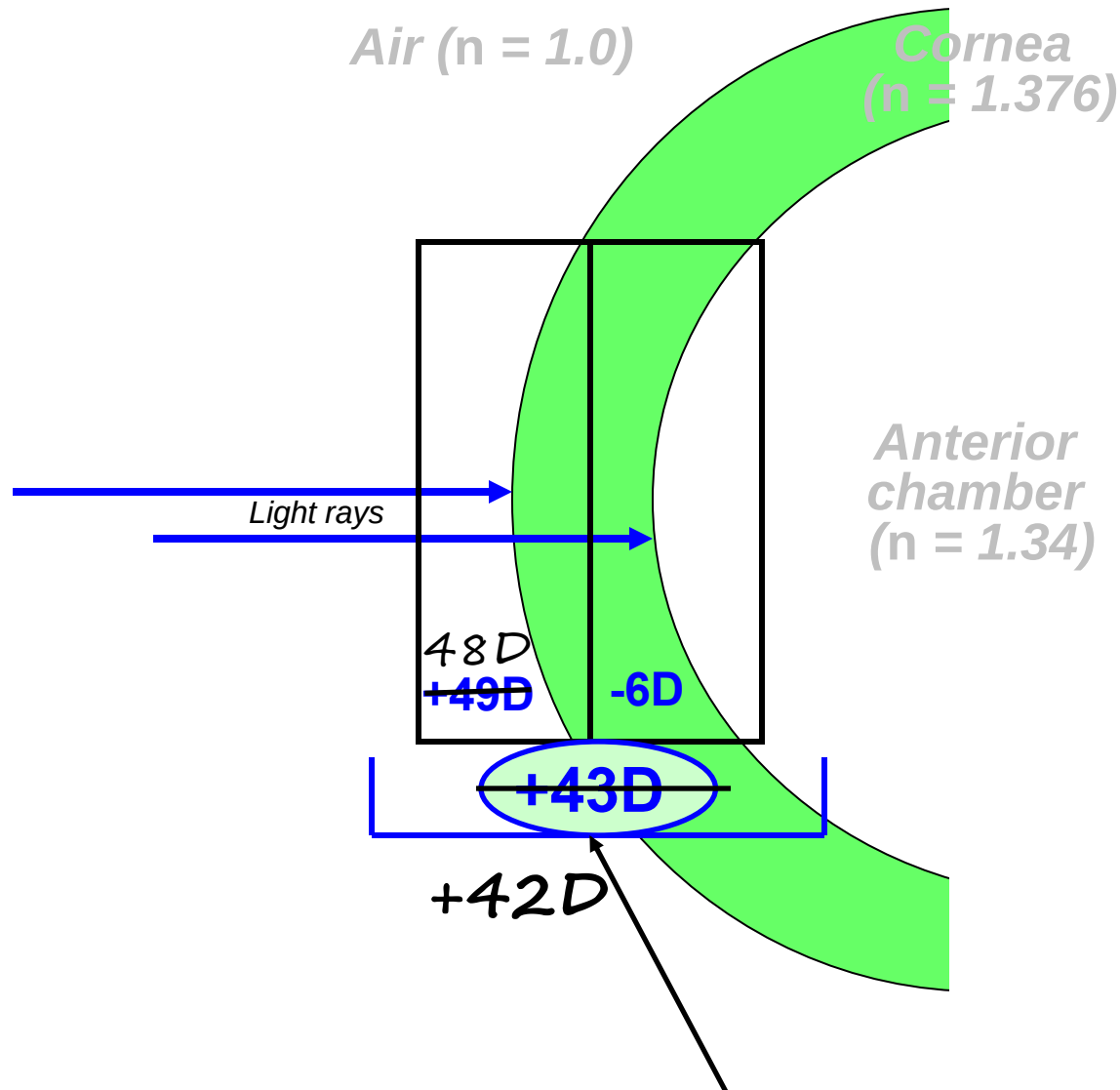
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Note: The BCSC *Clinical Optics* book uses these numbers. However...

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Corneal Optics

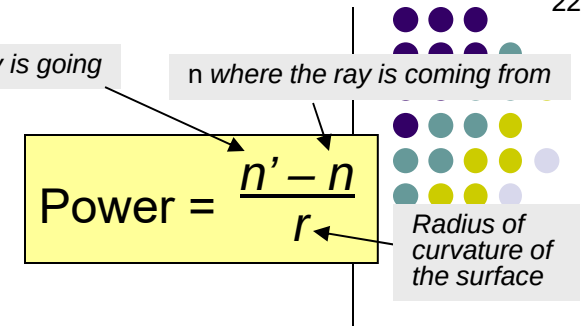


Note: The BCSC *Clinical Optics* book uses these numbers. However... The *Refractive Surgery* book assumes an anterior-corneal power of **48D**, which in turn results in an overall corneal power of **42D**. So don't be confused if you see these numbers.

The net result across the cornea is an overall power of about ~~43D~~ **42D**

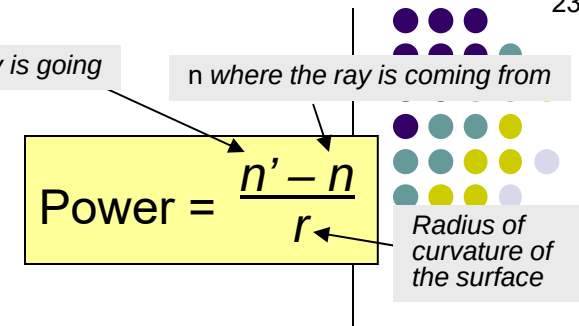
Corneal Optics

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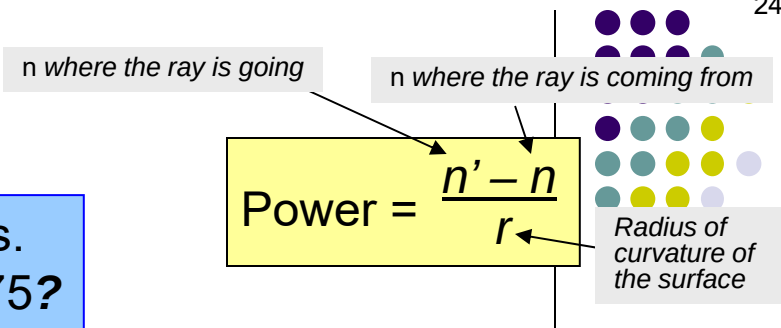
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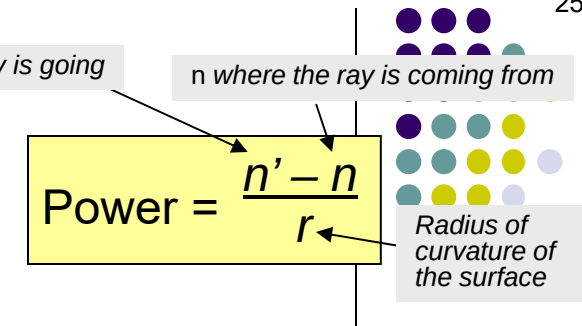


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$$n_{\text{cornea}} - \text{n of air } 1.0$$

Corneal Optics

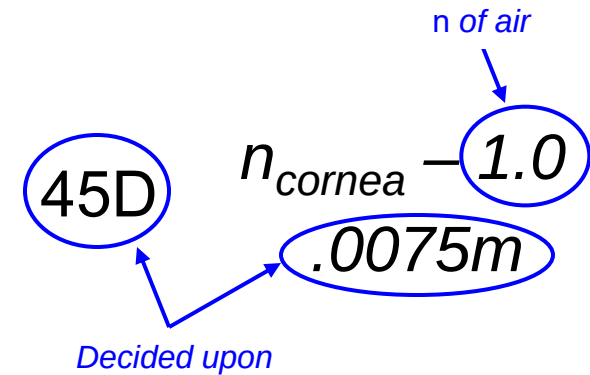
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Instead, it was decided to select a single n to capture overall corneal power. *Further, it was decided that an anterior radius-of-curvature of 7.5mm (0.0075m) should coincide with a total corneal power of 45D. (Who did all this deciding, I have no idea.)*

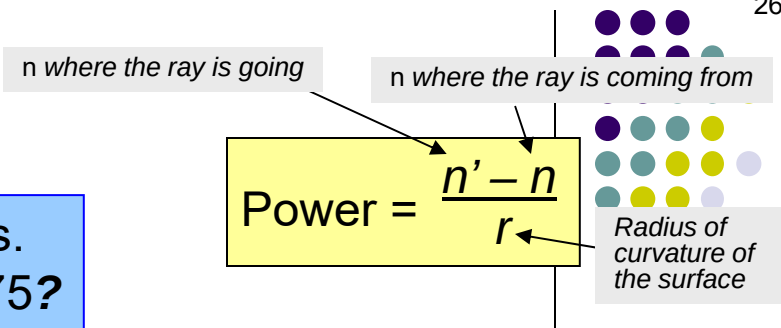


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$$45\text{D} = \frac{n_{\text{cornea}} - 1.0}{.0075\text{m}}$$

Labels in the diagram:

- 1.0 : n of air
- 45D : Decided upon

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Thus, assigning the cornea a single n of 1.3375 accounts for both its anterior (plus) and posterior (minus) effects, greatly simplifying calculations of net corneal power. However, 1.3375 does **not** accurately represent the 'true' refractive effect that corneal tissue has on light. In short, it is a **convenient fiction**.

n where the ray is going

n where the ray is coming from

$$\text{Power} = \frac{n' - n}{r}$$

Radius of curvature of the surface

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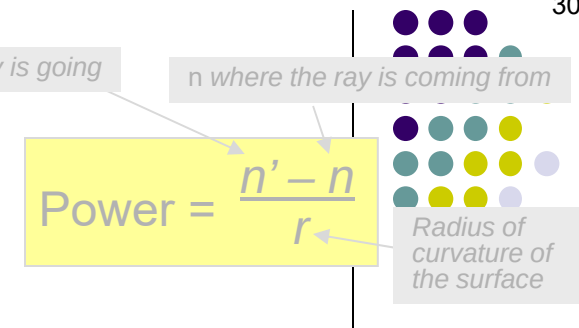
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$$D_{\text{cornea}} = \frac{1.3333 - 1.0}{.0075\text{m}}$$

Decided upon

n of air

$$D_{\text{cornea}} = 44.44$$

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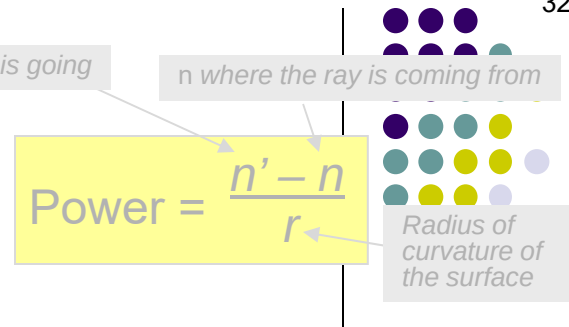
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The term $\frac{1.34 - 1.376}{r_{\text{Posterior}}}$ is circled in yellow with a red 'X' over it, and the text 'is proportional to' is written next to it.

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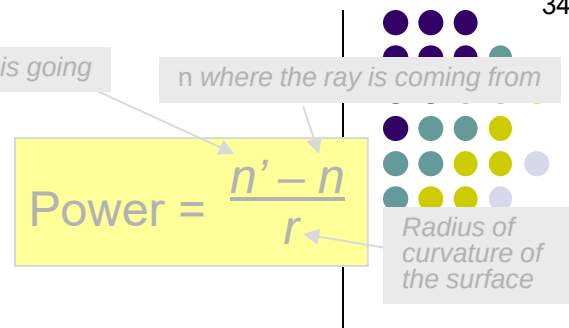
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So why is the cornea said to have an n of 1.3375?

*Determination of the **true** corneal power requires measurement of the posterior corneal curvature—a technically difficult and time-consuming task back in the day.*

$$\text{Power} = \frac{1.376 - 1.0}{r_{\text{Anterior}}} + \frac{1.34 - 1.376}{r_{\text{Posterior}}}$$

is proportional to

As stated previously, introduction of this *net corneal index of refraction* simplified the process of corneal power calculations. However, one of the things that makes this process possible is the

Recall also that keratometers determine central corneal power by assessing midperipheral corneal power, and then extrapolating based on an assumed relationship between midperipheral- and central corneal powers. Both keratoablative *and* keratoincisional (ie, RK) surgeries decouple this midperipheral power<-> central power relationship. This is another important source of error leading to refractive surprise after cataract surgery.

after the radius-of-curvature of the anterior corneal surface without affecting the posterior one, thereby de-coupling (un-correlating) the two curvatures. So what's the problem? Recall that some devices determine corneal power by estimating the cornea's anterior curvature and then extrapolating based on the net corneal index of refraction and the assumed constant relationship between anterior and posterior radius-of-curvature. However, s/p keratoablative surgery this relationship no longer holds, and thus the extrapolated corneal power will be **way** off. This is why post-keratoablative patients who undergo cataract surgery are at risk for a so-called 'refractive surprise'—their corneal measurements indicate one power, but the **actual** power is very different.

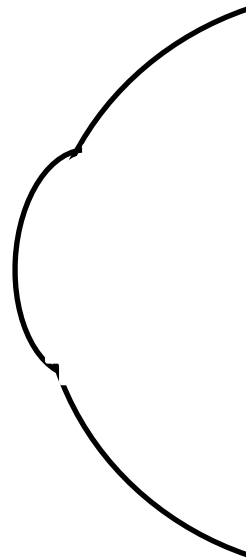
*However, 1.3375 does **not** accurately represent the 'true' refractive effect that corneal tissue has on light. In short, it is a **convenient fiction**.*



Q

Corneal Optics

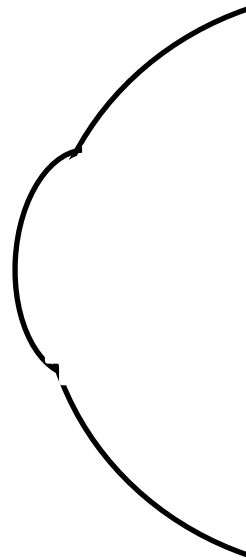
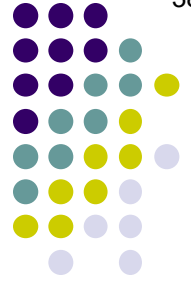
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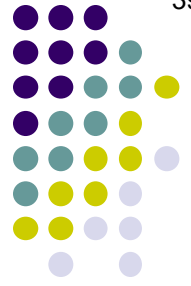


A

Corneal Optics

- *What one word best describes the overall shape of the cornea?* **Prolate**

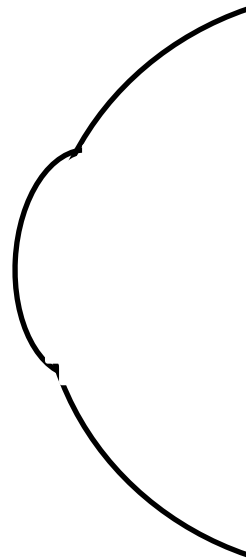


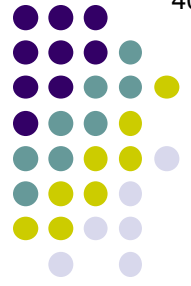


Q

Corneal Optics

- *What one word best describes the overall shape of the cornea? Prolate*
- *What does prolate indicate about the cornea's shape?*

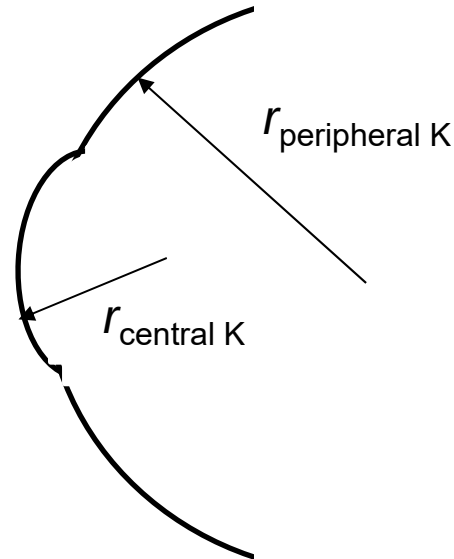


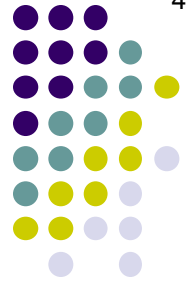


A

Corneal Optics

- *What one word best describes the overall shape of the cornea?* **Prolate**
- *What does prolate indicate about the cornea's shape?* **That the central cornea is steeper (ie, has a shorter radius of curvature) than the peripheral corneal**



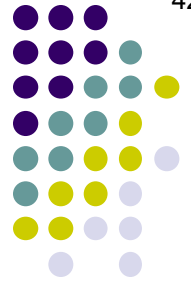


Q

Corneal Optics

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- *What does prolate indicate about the cornea's shape? That the central cornea is steeper (ie, has a shorter radius of curvature) than the peripheral corneal*

If the opposite relationship holds true--ie, if the central cornea is flatter than the peripheral--what word describes that shape?



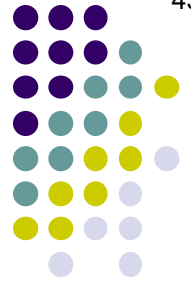
A

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Oblate



Q

Corneal Optics

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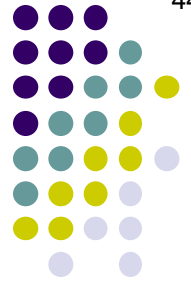
If the opposite relationship holds true--ie, if the central cornea is flatter than the peripheral--what word describes that shape?

Oblate

Under what (quite common) circumstance might one encounter an oblate cornea?

Q/A

Corneal Optics

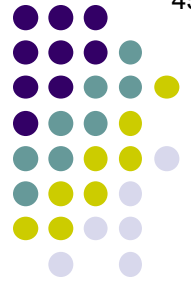


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Oblate

Under what (quite common) circumstance might one encounter an oblate cornea?
In an eye that is s/p keratorefractive surgery for



A

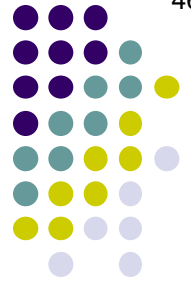
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If the opposite relationship holds true--ie, if the central cornea is flatter than the peripheral--what word describes that shape?

Oblate

Under what (quite common) circumstance might one encounter an oblate cornea?
In an eye that is s/p keratorefractive surgery for myopia



Q

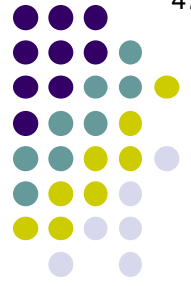
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Oblate

What if the entire cornea has the same steepness/radius of curvature--what word describes that shape?



A

Corneal Optics

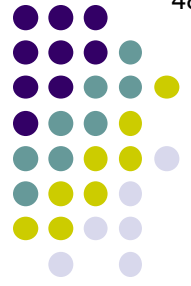
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Oblate

What if the entire cornea has the same steepness/radius of curvature--what word describes that shape?

Spherical



Q

Corneal Optics

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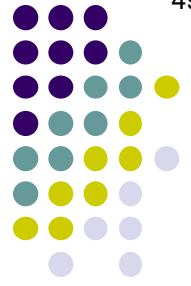
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Oblate

What if the entire cornea has the same steepness/radius of curvature--what word describes that shape?

Spherical

Does the human cornea ever take on a spherical shape?



A

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Oblate

What if the entire cornea has the same steepness/radius of curvature--what word describes that shape?

Spherical

Does the human cornea ever take on a spherical shape?

Essentially never



Q

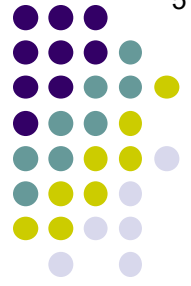
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What term quantifies the degree of difference between the curvature of the central cornea vs that of the corneal periphery?
steeper
curvature

If the opposite relationship exists--what word describes that shape?
 Oblate

heral--

What if the entire cornea has the same steepness/radius of curvature--what word describes that shape?
 Spherical



A

Corneal Optics

- What one word best describes the overall shape of the cornea? Prolate
- What does prolate indicate about the cornea's steepness and curvature?

What term quantifies the degree of difference between the curvature of the central cornea vs that of the corneal periphery?

The **Q factor** (aka **Q value**)

If the opposite relationship exists--what word describes that shape?
Oblate

heral--

What if the entire cornea has the same steepness/radius of curvature--what word describes that shape?
Spherical



Q

Corneal Optics

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If the opposite relationship exists, what word describes that shape?
Oblate

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Spherical

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A

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Spherical



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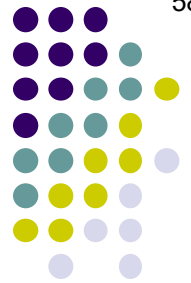
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Q

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Spherical

What is the unit of measurement for the Q factor?



A

Corneal Optics

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If the opposite relationship exists, what word describes that shape?

Oblate

What if the entire cornea has the same steepness/radius of curvature--what word describes that shape?

Spherical

What is the unit of measurement for the Q factor?
It has none—it is a dimensionless number



Q

Corneal Optics

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cornea
steeper
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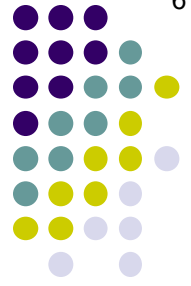
If the opposite re
what word descri
Oblate

heral--

What if the entire cornea has the same steepness/radius of curvature--what word describes that shape?

Spherical

What is the average Q factor/value for the human cornea?



A

Corneal Optics

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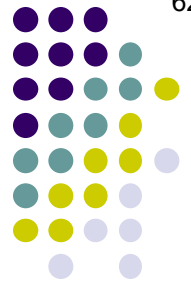
If the opposite re
what word descri
Oblate

heral--

What if the entire cornea has the same steepness/radius of curvature--what word describes that shape?

Spherical

What is the average Q factor/value for the human cornea?
-0.26



Q

Corneal Optics

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If the opposite re
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Spherical

In one word, what property of the cornea does the Q factor quantify?



A

Corneal Optics

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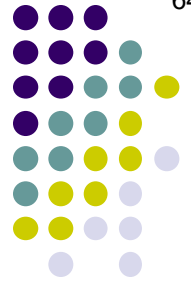
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If the opposite relationship exists, what word describes that shape?
Oblate

What if the entire cornea has the same steepness/radius of curvature--what word describes that shape?
Spherical

In one word, what property of the cornea does the Q factor quantify?
Corneal **asphericity** (in fact, it is often referred to as the 'asphericity Q factor')



Q

Corneal Optics

- *What one word best describes the overall shape of the cornea? Prolate*
- *What does prolate indicate about the cornea's shape? That the central cornea is steeper (ie, has a shorter radius of curvature) than the peripheral corneal*
- *Is any portion of the cornea spherical?*



A

Corneal Optics

- *What one word best describes the overall shape of the cornea?* **Prolate**
- *What does prolate indicate about the cornea's shape?* **That the central cornea is steeper (ie, has a shorter radius of curvature) than the peripheral corneal**
- *Is any portion of the cornea spherical?*
Yes, the central ~3mm closely approximates a spherical surface



Q

Corneal Optics

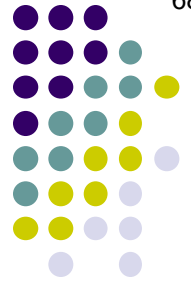
- *What one word best describes the overall shape of the cornea?* **Prolate**
- *What does the prolate nature of the human cornea mean for visual acuity?*
On balance, and with respect to visual acuity, is the prolate nature of the human cornea a good thing?
- *Is any portion of the cornea spherical?*
Yes, the central ~3mm closely approximates a spherical surface



A

Corneal Optics

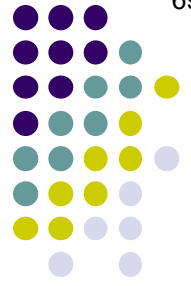
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On balance, and with respect to visual acuity, is the prolate nature of the human cornea a good thing?
Yes, definitely
- *Is any portion of the cornea spherical?*
Yes, the central ~3mm closely approximates a spherical surface



Q

Corneal Optics

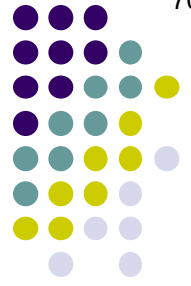
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- *What does the prolate nature of the human cornea mean for visual acuity?*
On balance, and with respect to visual acuity, is the prolate nature of the human cornea a good thing?
Yes, definitely
- *Why is the cornea steeper in the center?*
It seems like a spherical cornea would be better. Why is this not the case?
- *Is any portion of the cornea spherical?*
Yes, the central ~3mm closely approximates a spherical surface



A

Corneal Optics

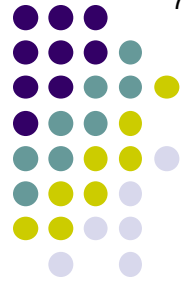
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- *What does the prolate nature of the human cornea mean for visual acuity?*
On balance, and with respect to visual acuity, is the prolate nature of the human cornea a good thing?
Yes, definitely
- *Why is a spherical cornea not better?*
It seems like a spherical cornea would be better. Why is this not the case?
Because a spherical refracting structure results in more spherical aberration
- *Is any portion of the cornea spherical?*
Yes, the central ~3mm closely approximates a spherical surface



Q

Corneal Optics

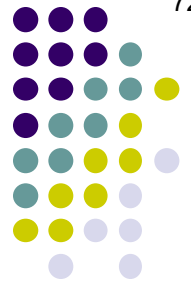
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- What does the prolate nature of the human cornea mean?
On balance, and with respect to visual acuity, is the prolate nature of the human cornea a good thing?
Yes, definitely
- Why is the cornea steeper in the center?
It seems like a spherical cornea would be better. Why is this not the case?
Because a spherical refracting structure results in more spherical aberration
- Is any portion of the cornea spherical?
Yes, the central portion of the cornea approximates a spherical surface
What is spherical aberration?



A

Corneal Optics

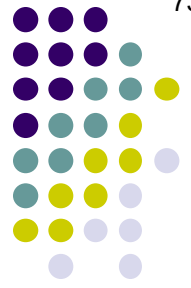
- What one word best describes the overall shape of the cornea? **Prolate**
- What does the prolate nature of the human cornea mean for visual acuity?
 On balance, and with respect to visual acuity, is the prolate nature of the human cornea a good thing?
 Yes, definitely
- What does the prolate nature of the human cornea mean for visual acuity?
 It seems like a spherical cornea would be better. Why is this not the case?
 Because a spherical refracting structure results in more **spherical aberration**
- Is any portion of the cornea spherical?
 Yes, the central portion of the cornea is spherical.
 What is spherical aberration?
 See slide-set **RS6** for a discussion of this important topic
- How does the prolate shape of the cornea approximate a spherical surface?



Q

Corneal Optics

- *What one word best describes the overall shape of the cornea?* **Prolate**
- *What does prolate indicate about the cornea's shape?* **That the central cornea is steeper (ie, has a shorter radius of curvature) than the peripheral corneal**
- *Is any portion of the cornea spherical?*
Yes, the central ~3mm closely approximates a spherical surface
- *Because the central cornea is steeper than the periphery, it has more power. How much more?*

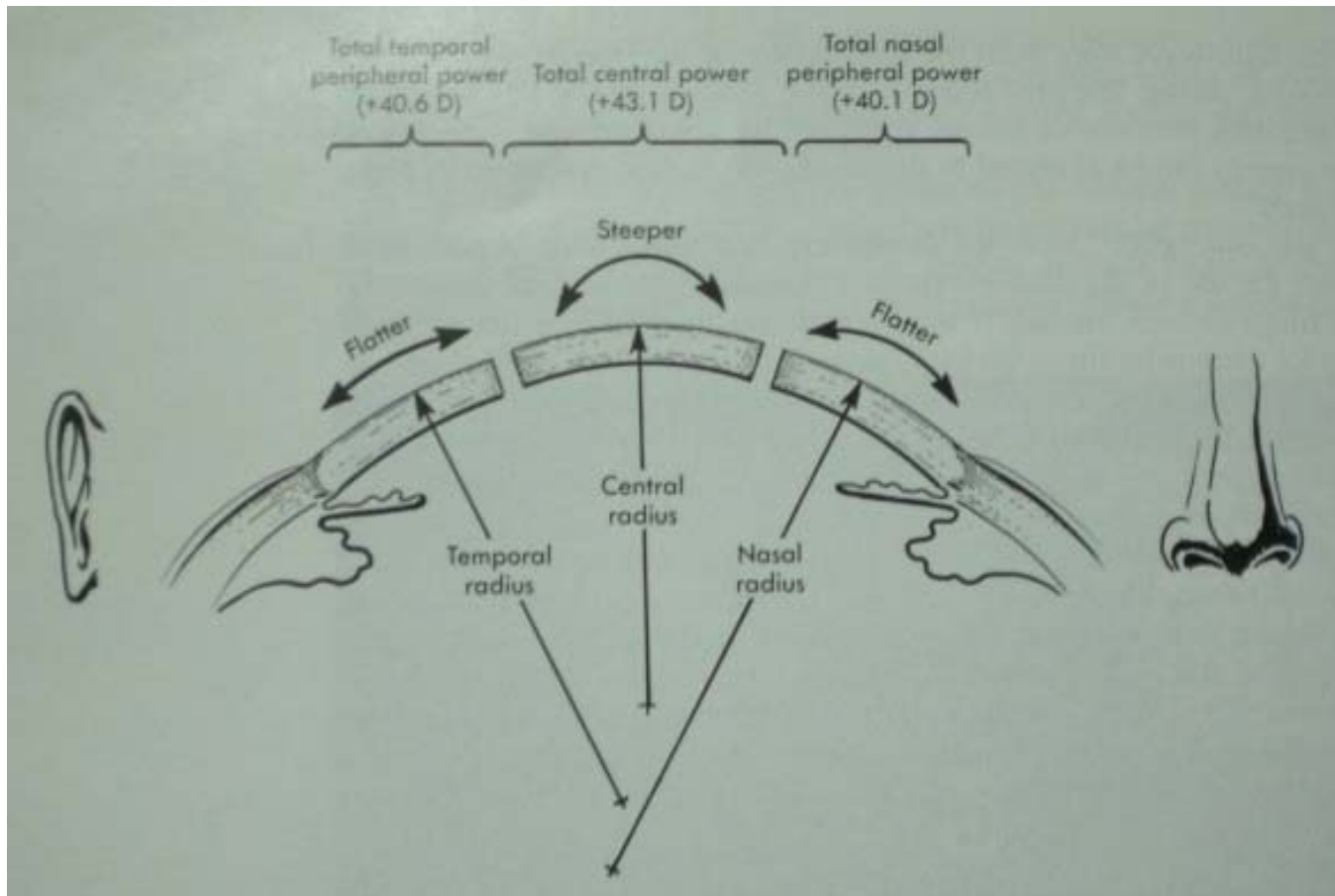


A

Corneal Optics

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Yes, the central ~3mm closely approximates a spherical surface
- *Because the central cornea is steeper than the periphery, it has more power. How much more?*
About 3-4 diopters more

Corneal Optics



Power differential of central vs peripheral prolate cornea

Q

Corneal Optics

- *What are the three categories of technology for determining central corneal power?*
 - ?
 - ?
 - ?

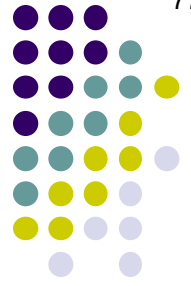


A

Corneal Optics

- *What are the three categories of technology for determining central corneal power?*
 - Keratometry
 - Corneal topography
 - Corneal tomography





Q

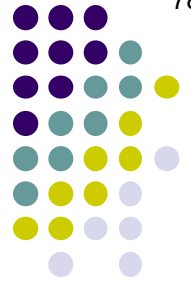
Corneal Optics

- *What are the three categories of technology for determining central corneal power?*
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How does a keratometer work?

A

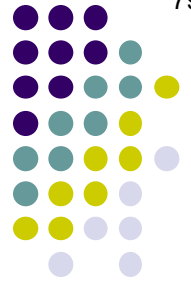
Corneal Optics



- *What are the three categories of technology for determining central corneal power?*
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How does a keratometer work?

By producing a reflection of an object of known size off the cornea (technically, off the tear film) from a known distance. By comparing the size of the resulting image to the size of the object, an estimate of corneal curvature can be calculated.



Q

Corneal Optics

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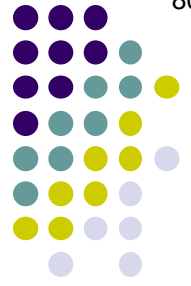
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So it measures central corneal curvature/power directly?

A

Corneal Optics



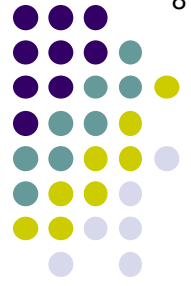
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No. It assesses the rate of curvature at 4 points in the paracentral zone of the cornea; ie, along a circle 3-4 mm in diameter centered on the corneal apex. Then, based on the assumptions discussed earlier in this slide-set, the central power of the cornea is estimated.



Q

Corneal Optics

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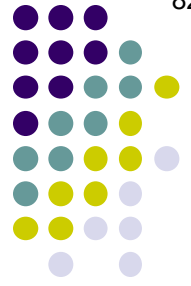
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Does keratometry measure the curvature of the posterior corneal surface?

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Corneal Optics



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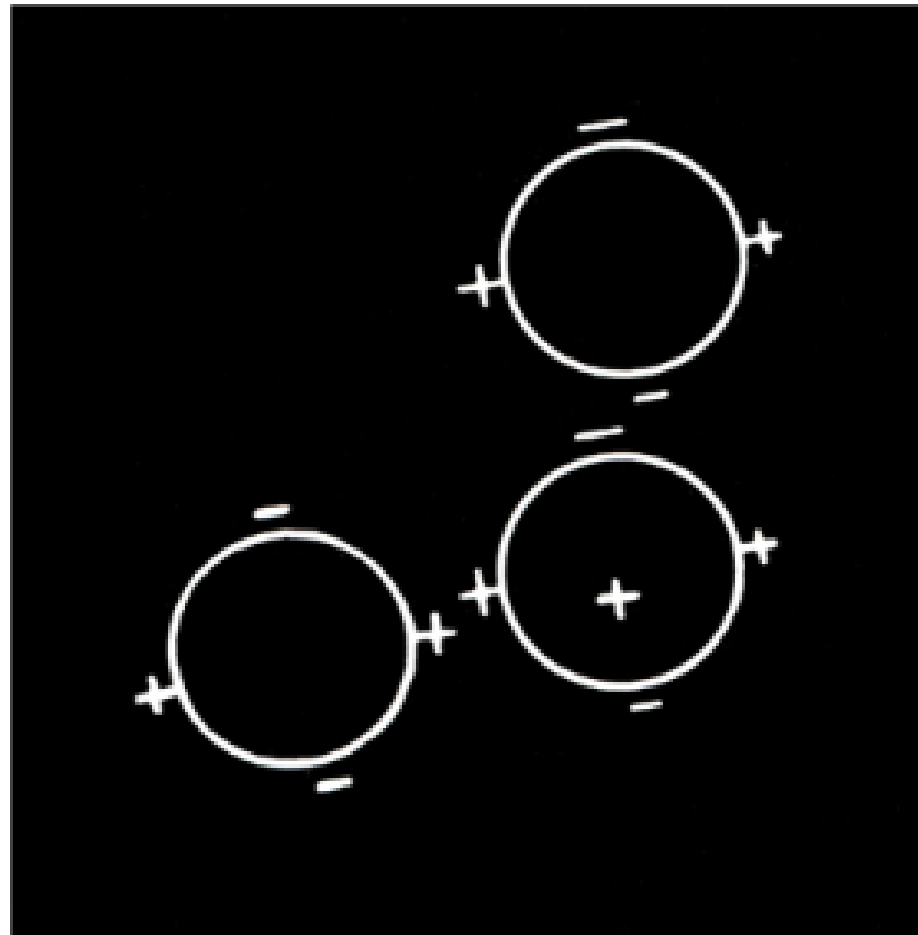
No

Corneal Optics

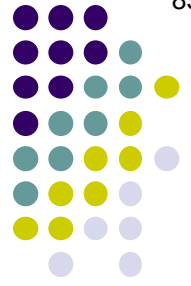


Keratometer (B&L type)

Corneal Optics



Keratometry image (B&L type)



Q

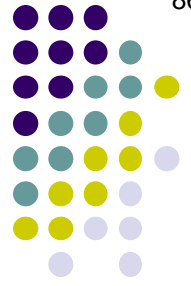
Corneal Optics

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How does a corneal topographer work?

A

Corneal Optics

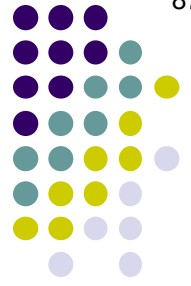


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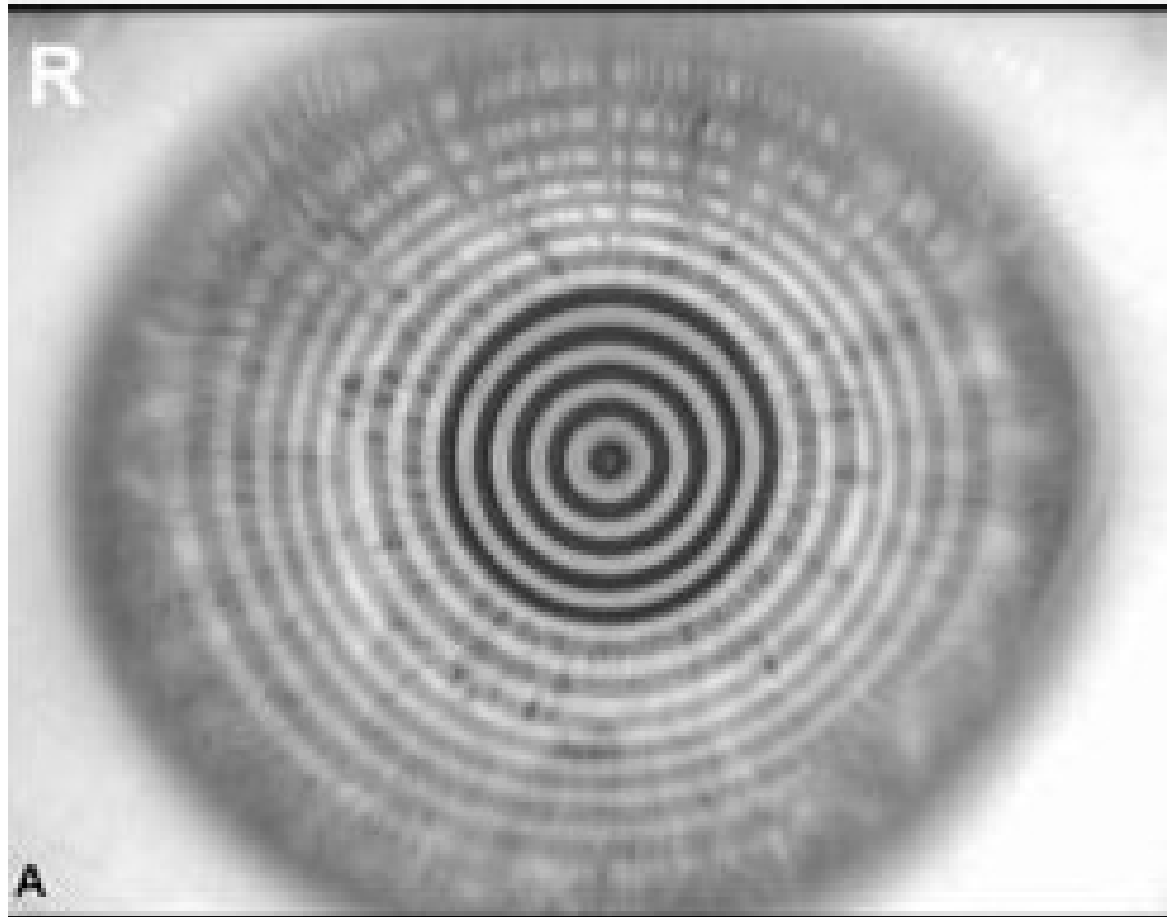
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Corneal Optics



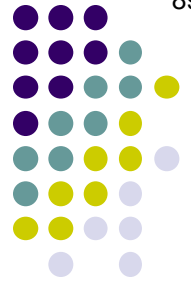
Corneal Placido-disk topography: Device



Corneal Placido-disk topography: Normal mires

A

Corneal Optics

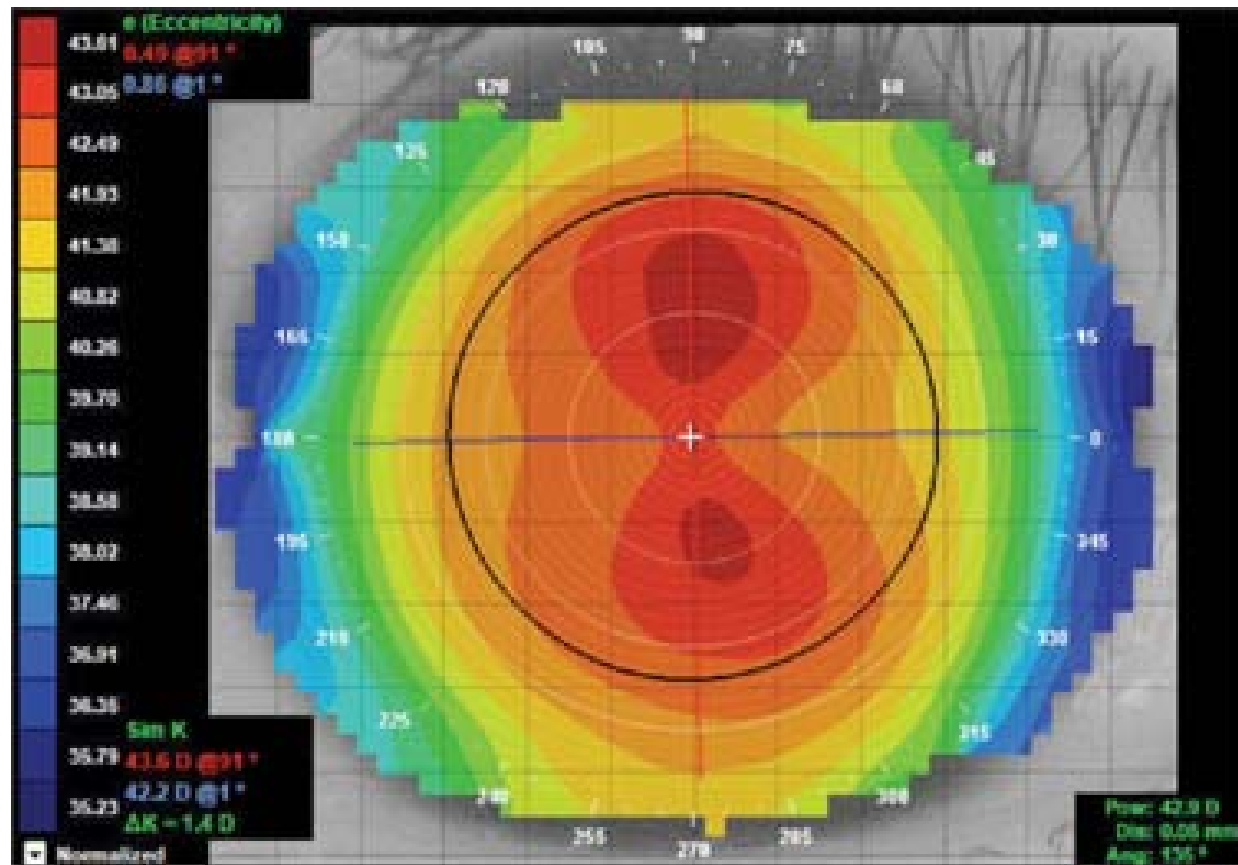


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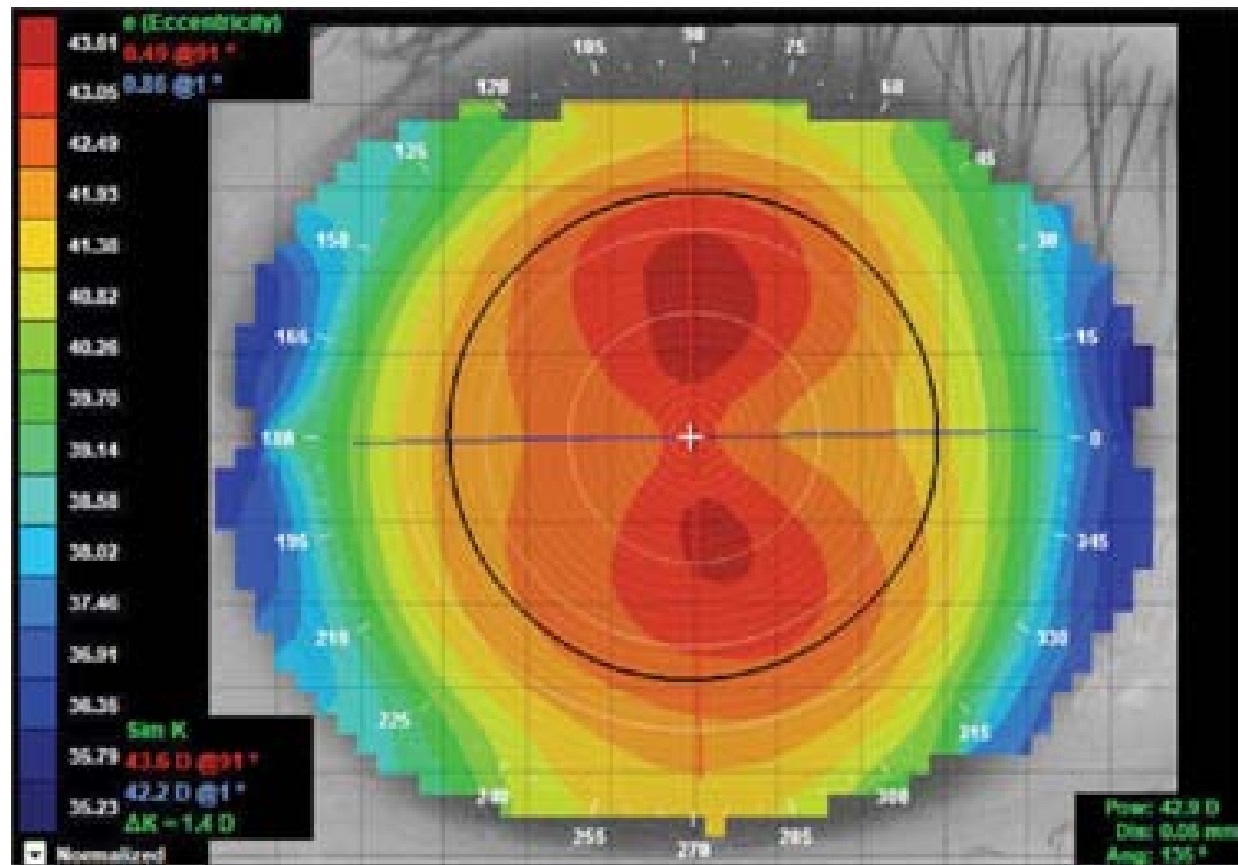
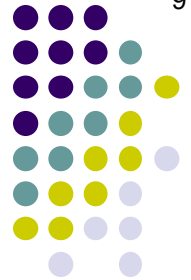
Corneal Optics



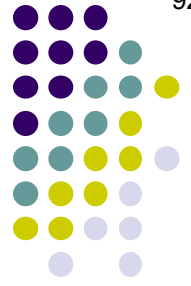
Corneal Placido-disk topography: Color map demonstrating...

Corneal Optics

91



Corneal Placido-disk topography: Color map demonstrating...with-the-rule astigmatism



Q

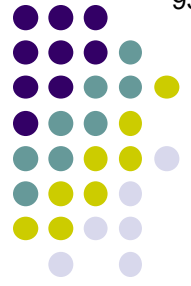
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In doing this, does corneal topography directly measure central corneal curvature/power?



A

Corneal Optics

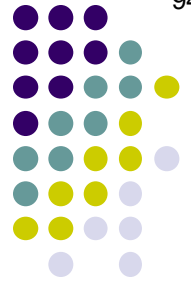
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Q

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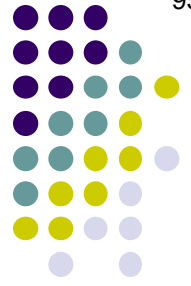
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Does topography measure the curvature/power of the posterior corneal surface?



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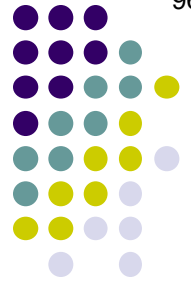
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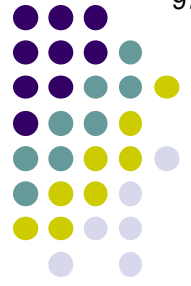
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Does topography measure the curvature/power of the posterior corneal surface?

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Does topography, of itself, provide adequate information to perform keratorefractive surgery?



A

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Provided it is interpreted properly, yes



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Why this qualifier? How might interpretation be compromised?

curvature/power?

topometer, topography still

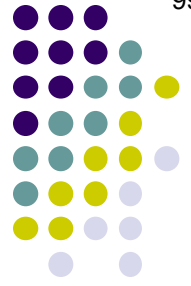
surface?

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The acquisition of topography data is susceptible to artifacts which can result in inaccurate mapping

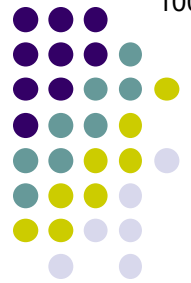
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--Artifacts secondary to...

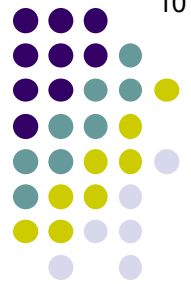
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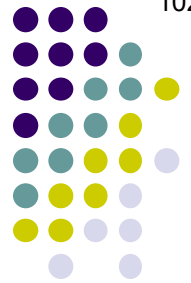
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Like keratom...
concentric ri...
the distance...
creates a co...

How might tear film deficiencies mess up topography?

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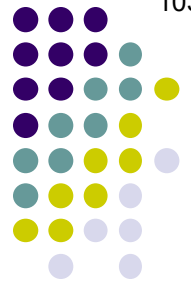
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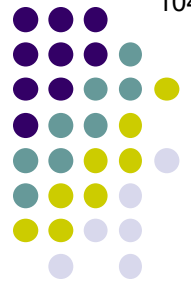
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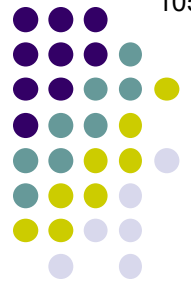
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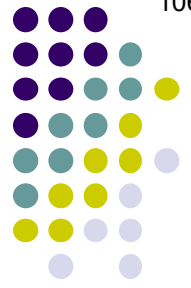
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Q/A

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--?

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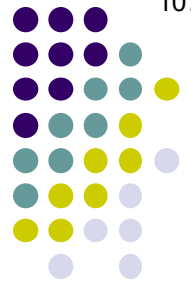
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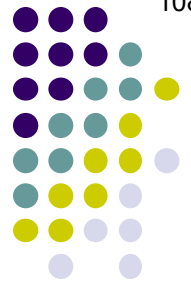
- The **visual** axis
- The **optical** or (**pupillary**) axis
- The **topographer** axis (aka the **vertex normal**)

What sorts of artifacts interfere with interpretation of corneal topography?

- Artifacts secondary to...abnormalities/deficiencies of the tear film
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Like keratometry, corneal topography is reflection-based. However, topographers reflect a set of

Why is alignment of the topographer an issue?

Because there are three different 'axes' involved. They are defined as:

--The **visual** axis: The line connecting the...?

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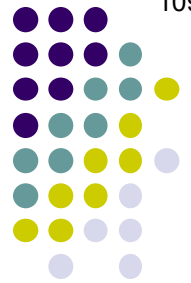
What sorts of artifacts interfere with interpretation of corneal topography?

--Artifacts secondary to...abnormalities/deficiencies of the tear film

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Does topography, of itself, provide adequate information to perform keratorefractive surgery?

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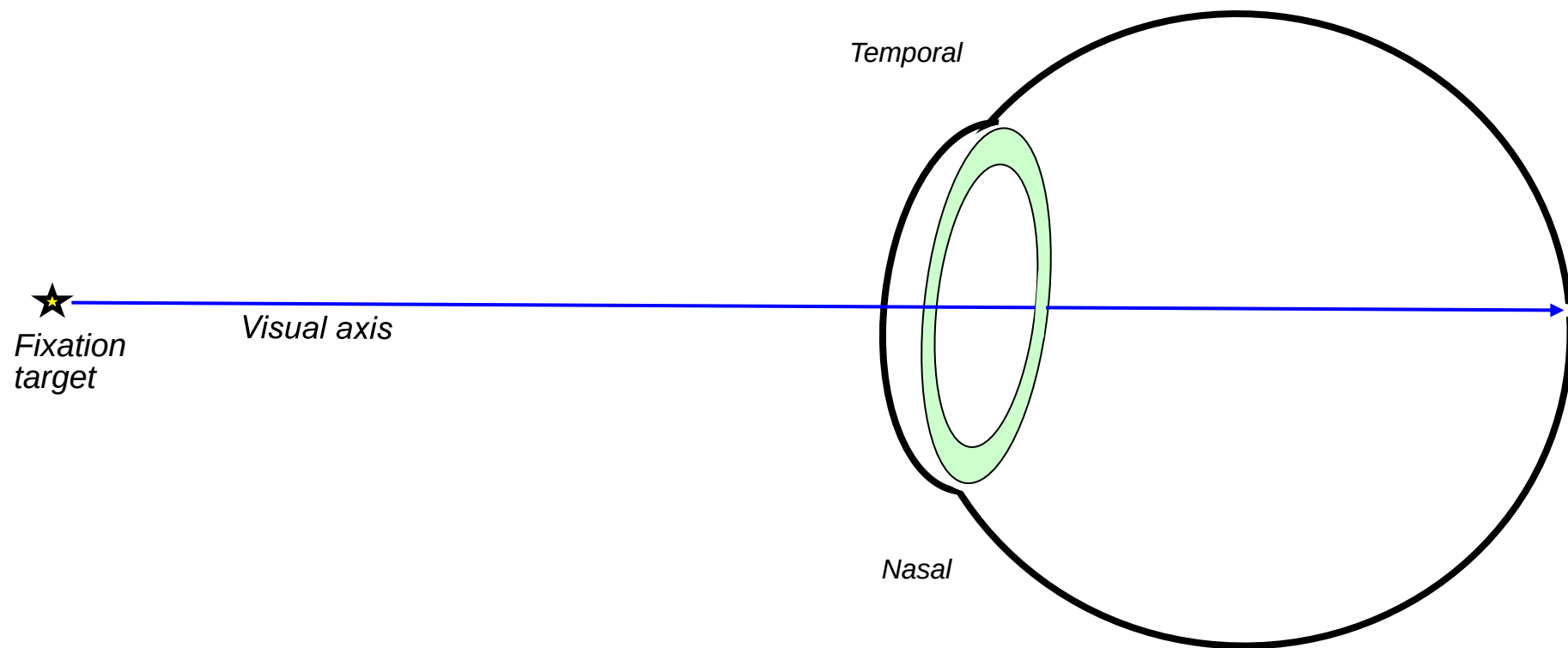
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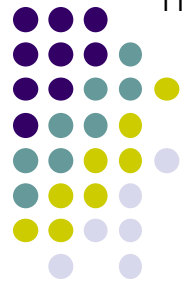


(OD, viewed from above)



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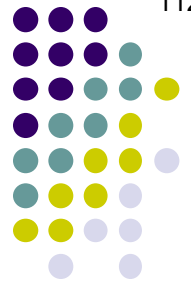
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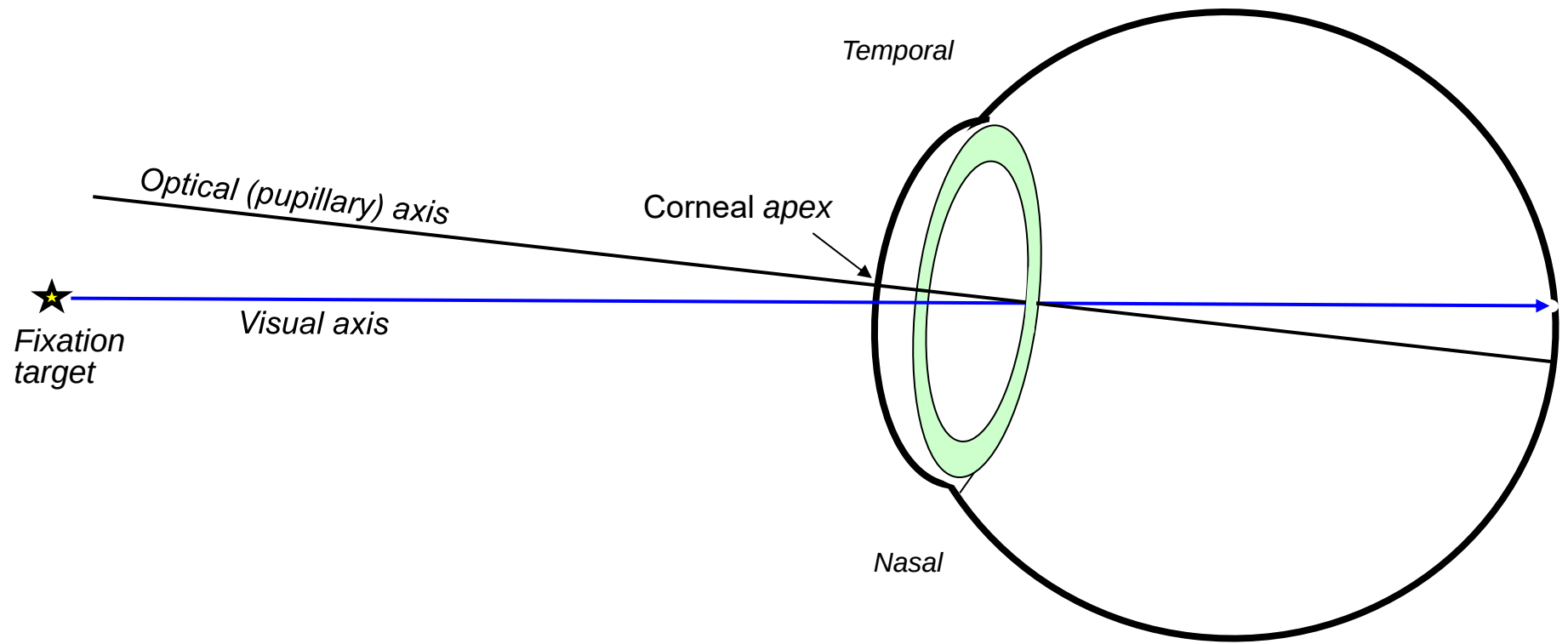
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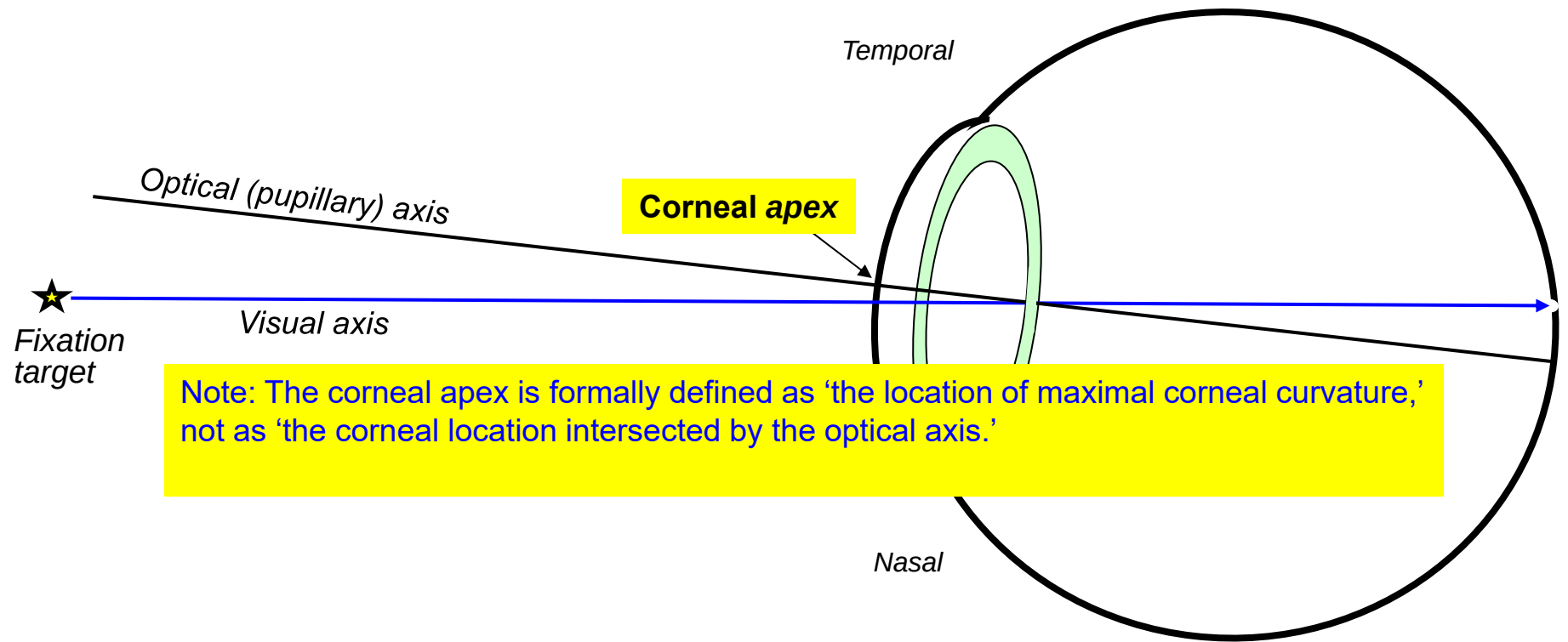
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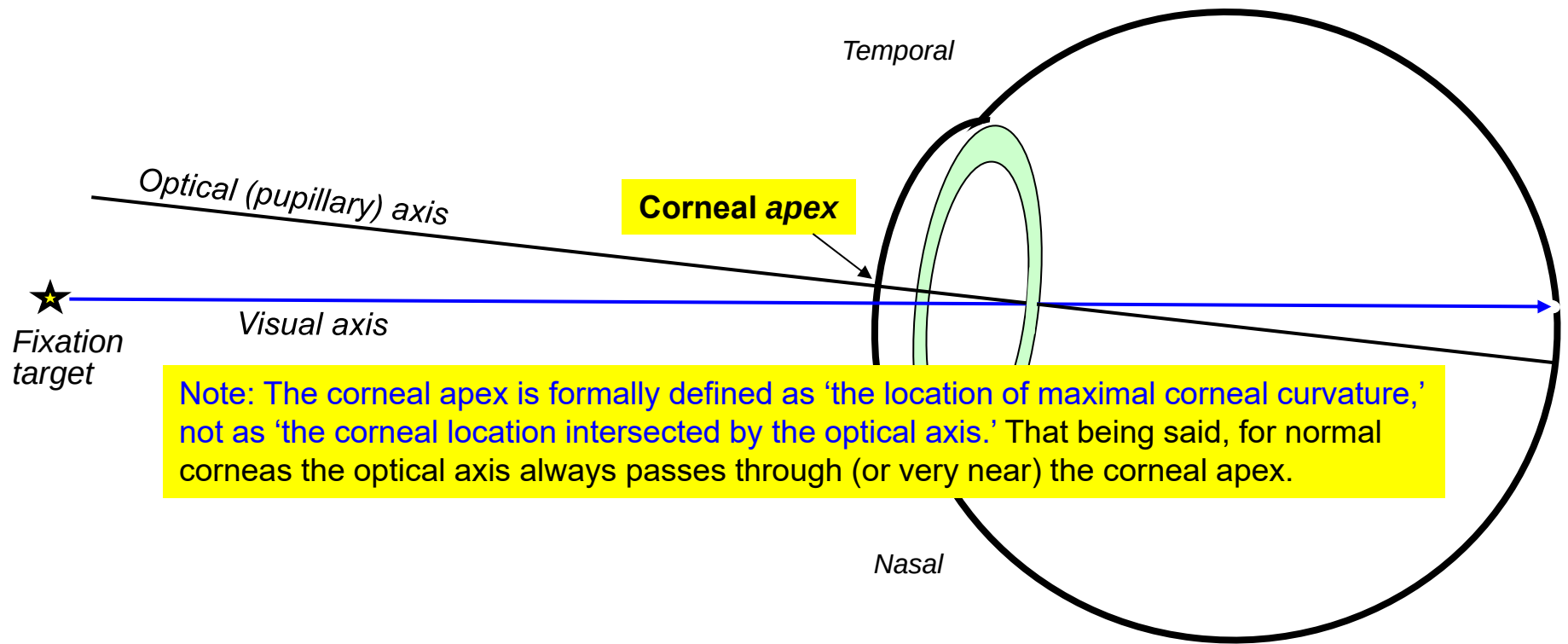
Corneal Optics

(OD, viewed from above)



Corneal Optics

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Note: The corneal apex is formally defined as 'the location of maximal corneal curvature,' not as 'the corneal location intersected by the optical axis.' That being said, for normal corneas the optical axis always passes through (or very near) the corneal apex.

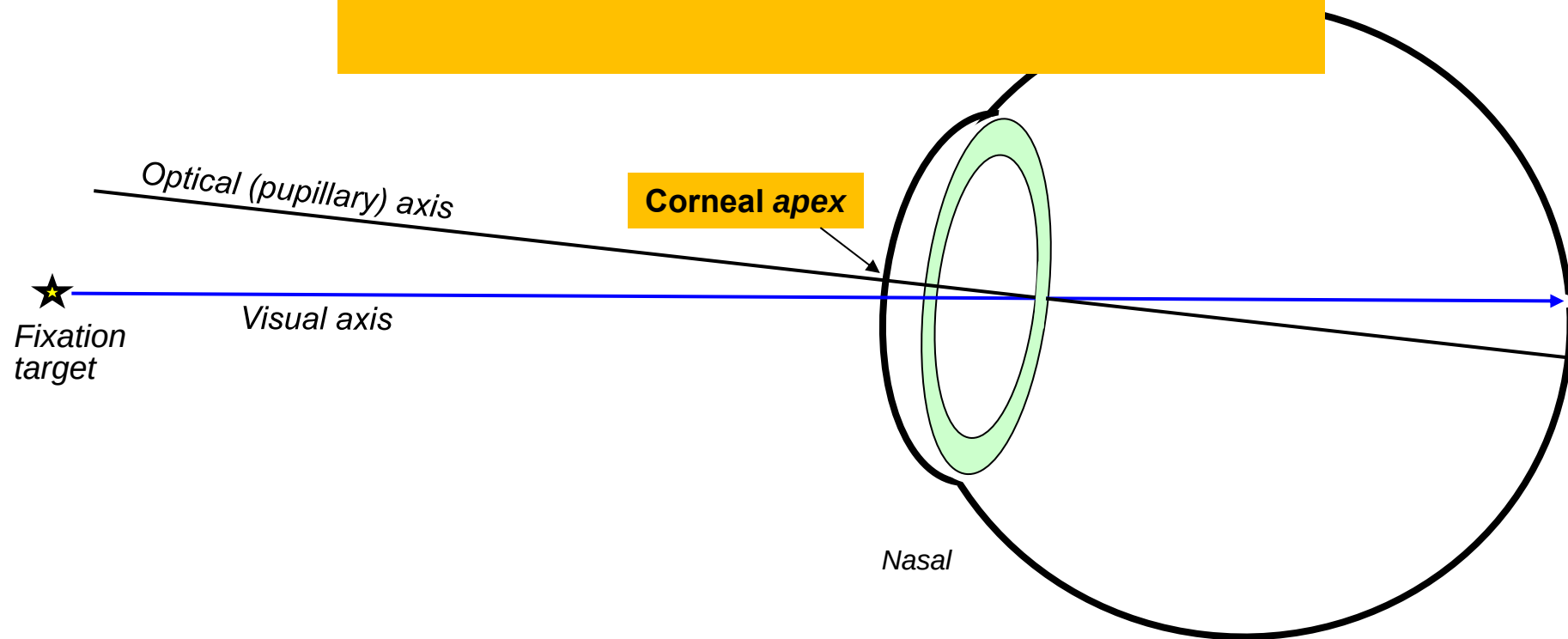
Q

Corneal Optics



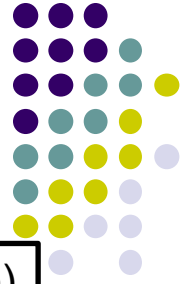
How can one readily identify the corneal apex in the clinic?

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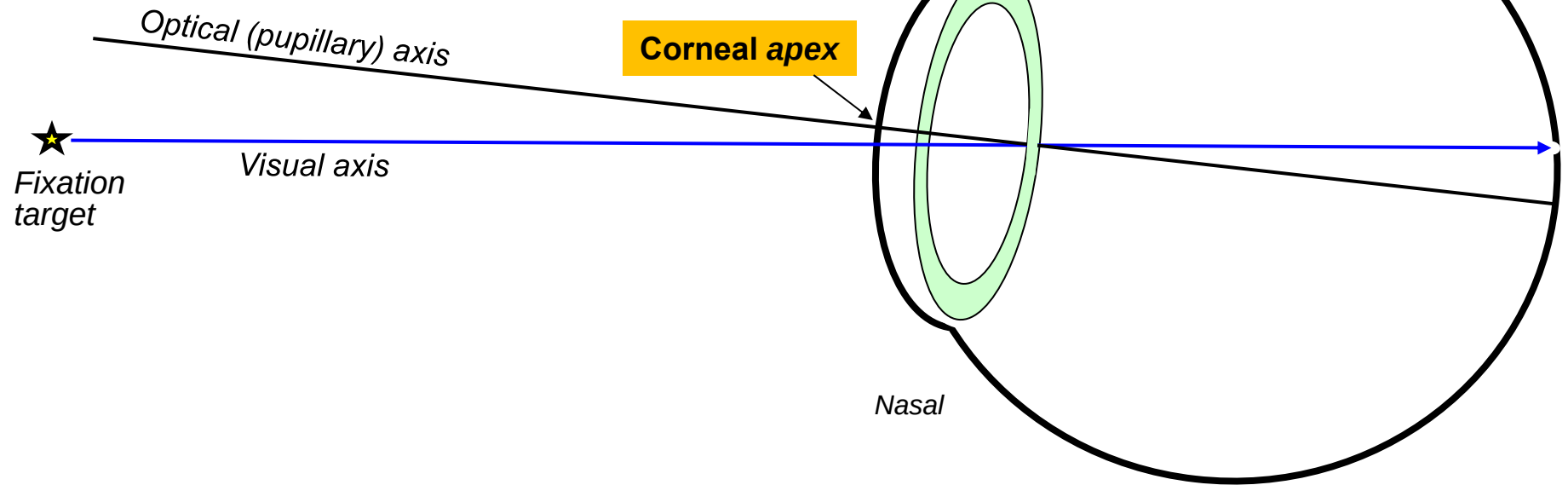


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Corneal Optics

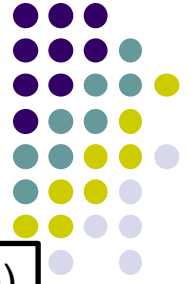


How can one readily identify the corneal apex in the clinic?
When you align a light-source with your line of sight, then align the location of the corneal light reflex with the center of the pt's pupil, that corneal location is the corneal apex

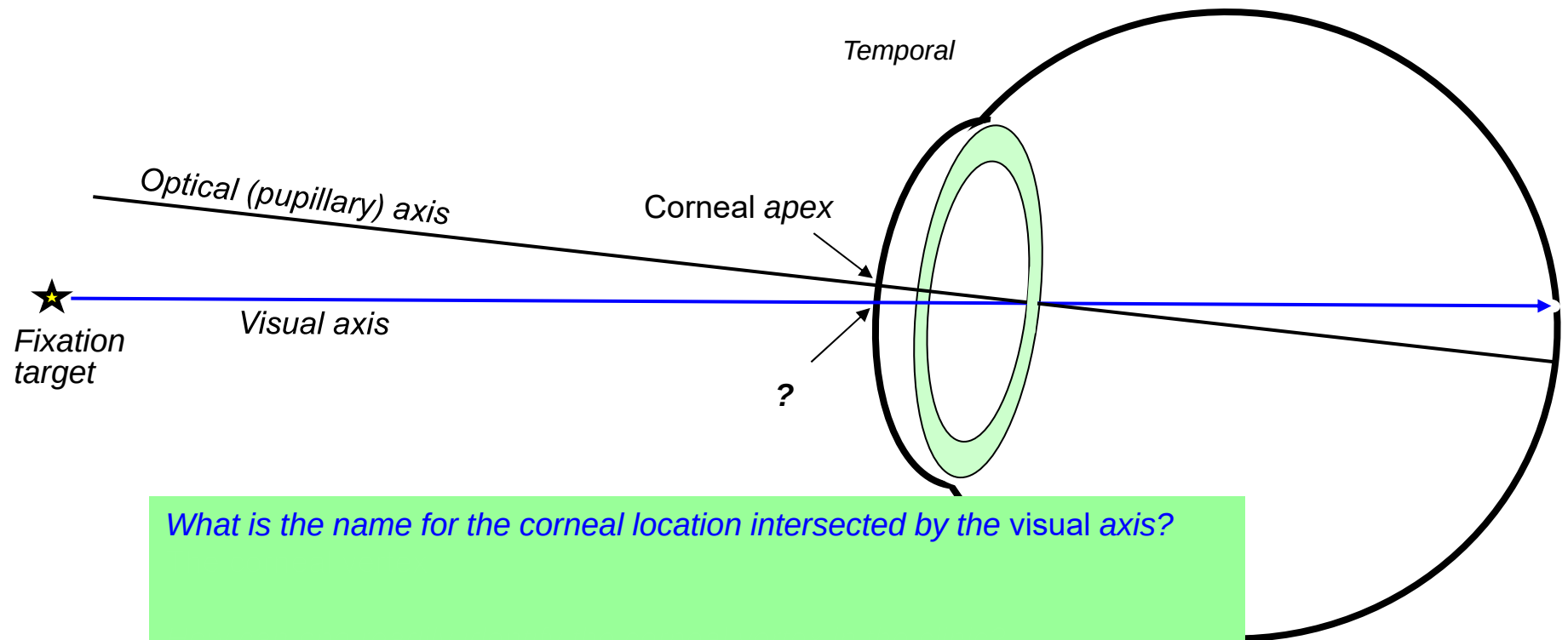


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Corneal Optics



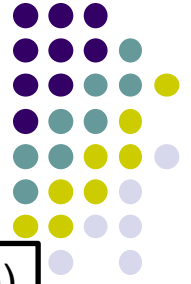
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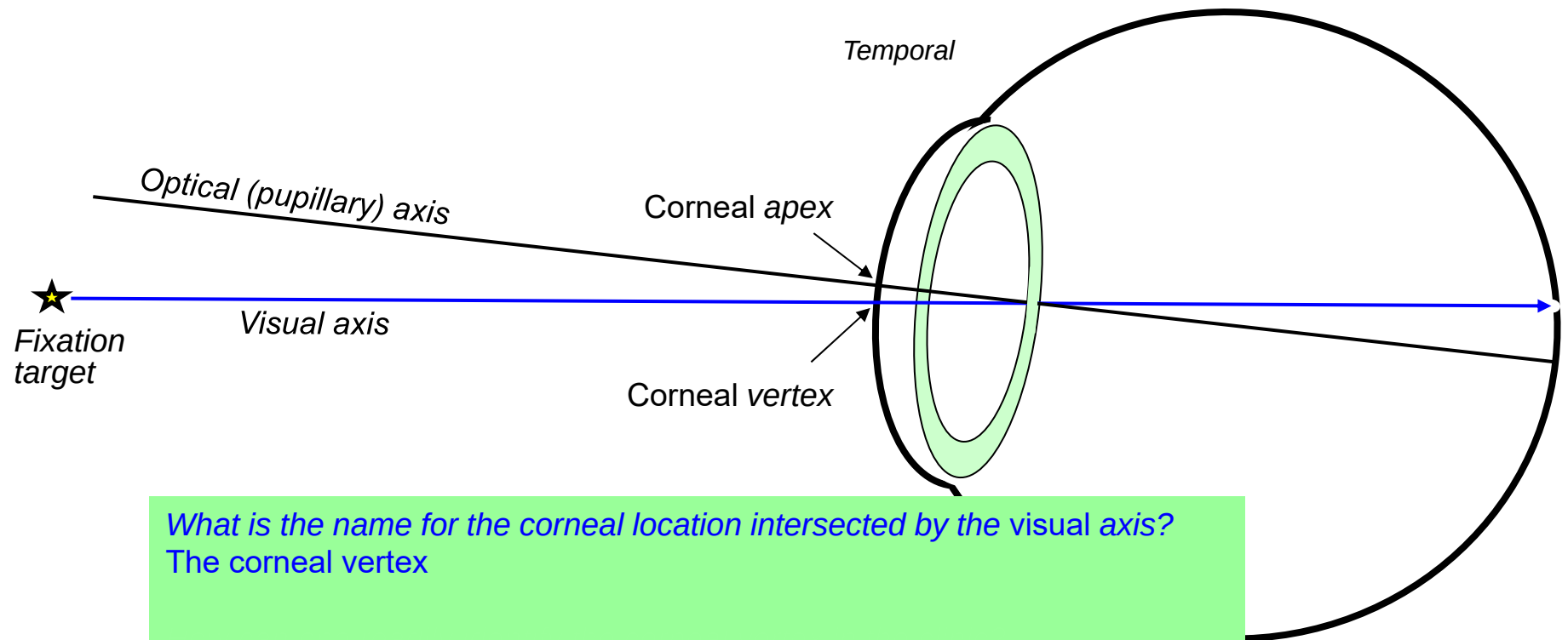
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A

Corneal Optics

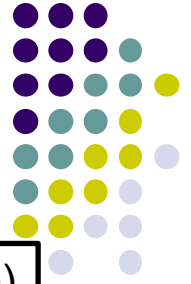


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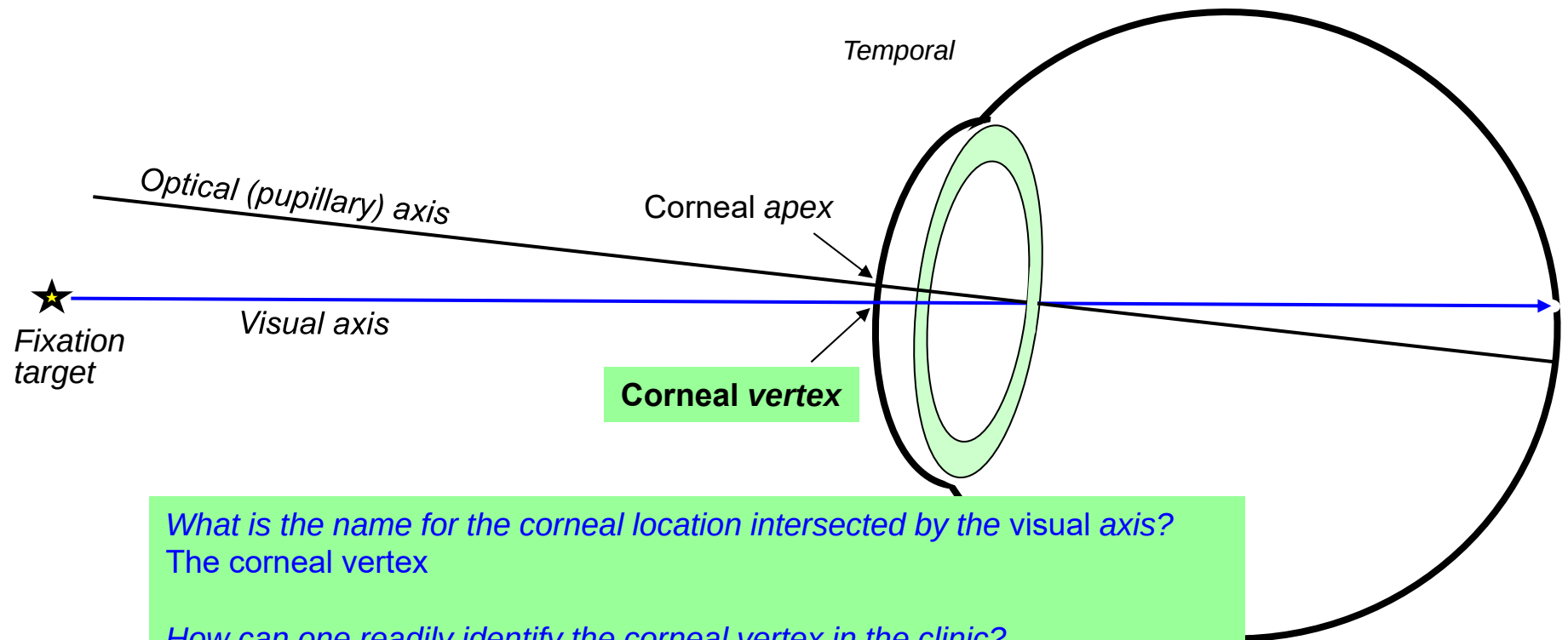


Q

Corneal Optics



(OD, viewed from above)

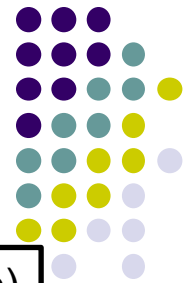


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The corneal vertex

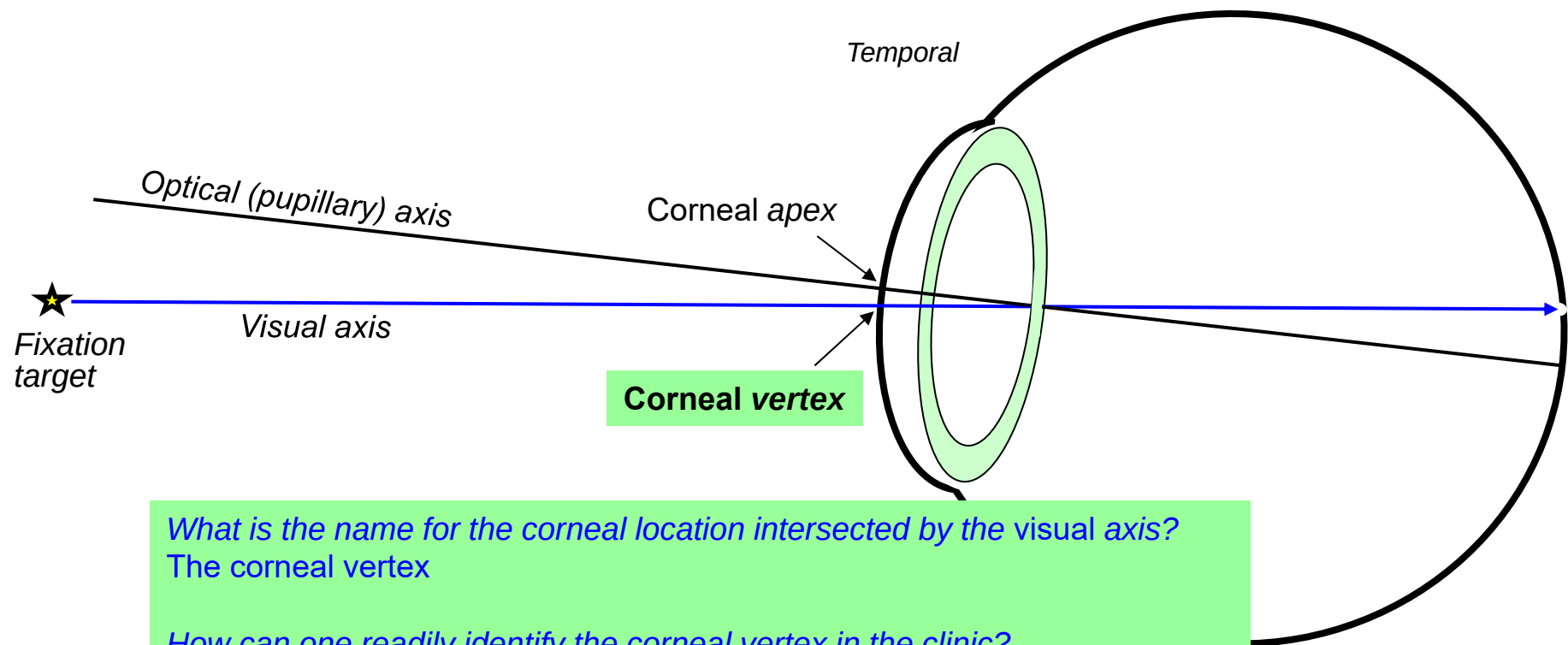
How can one readily identify the corneal vertex in the clinic?

A

Corneal Optics



(OD, viewed from above)

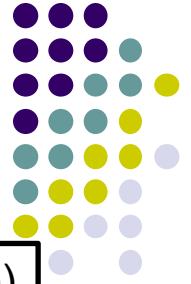


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How can one readily identify the corneal vertex in the clinic?
When you align a light-source with your line of sight, then ask the pt to look at the light, the location of the corneal light reflex is the corneal vertex

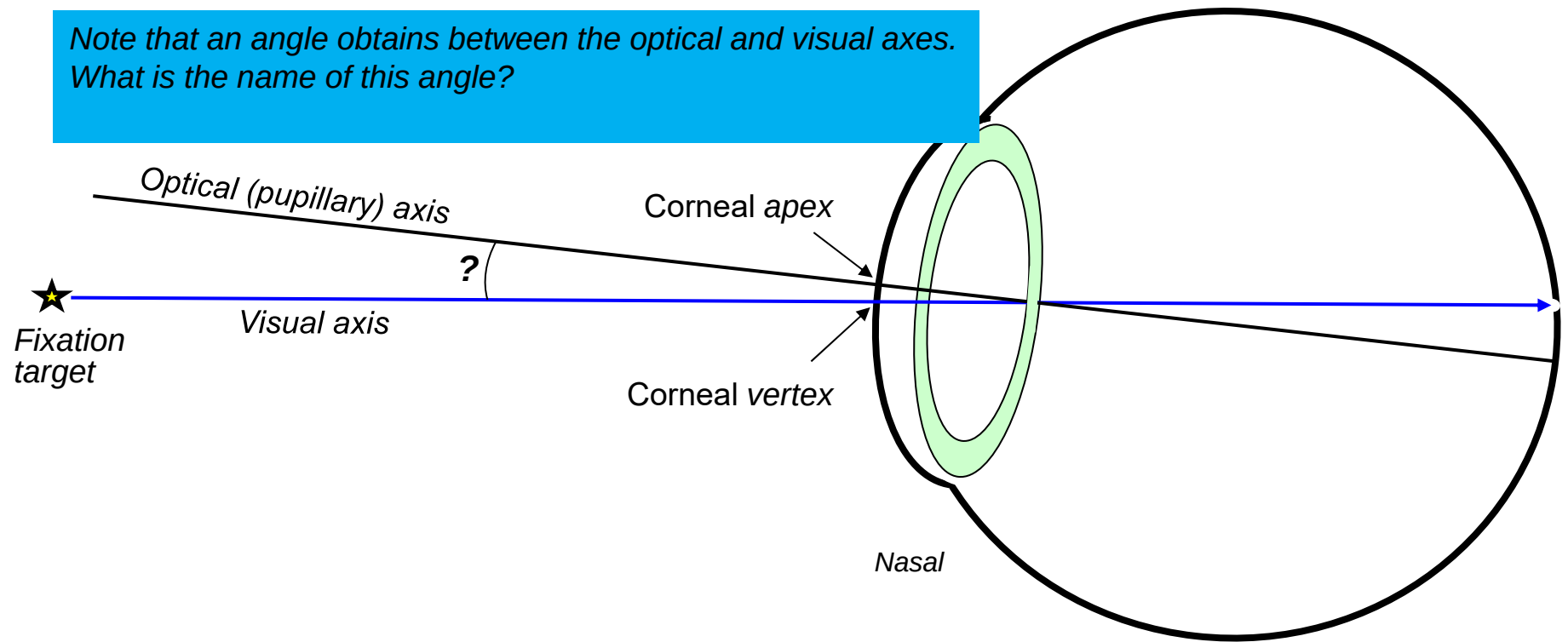
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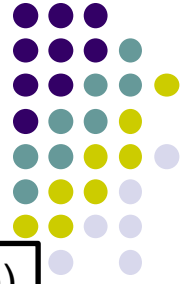
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Note that an angle obtains between the optical and visual axes.
What is the name of this angle?



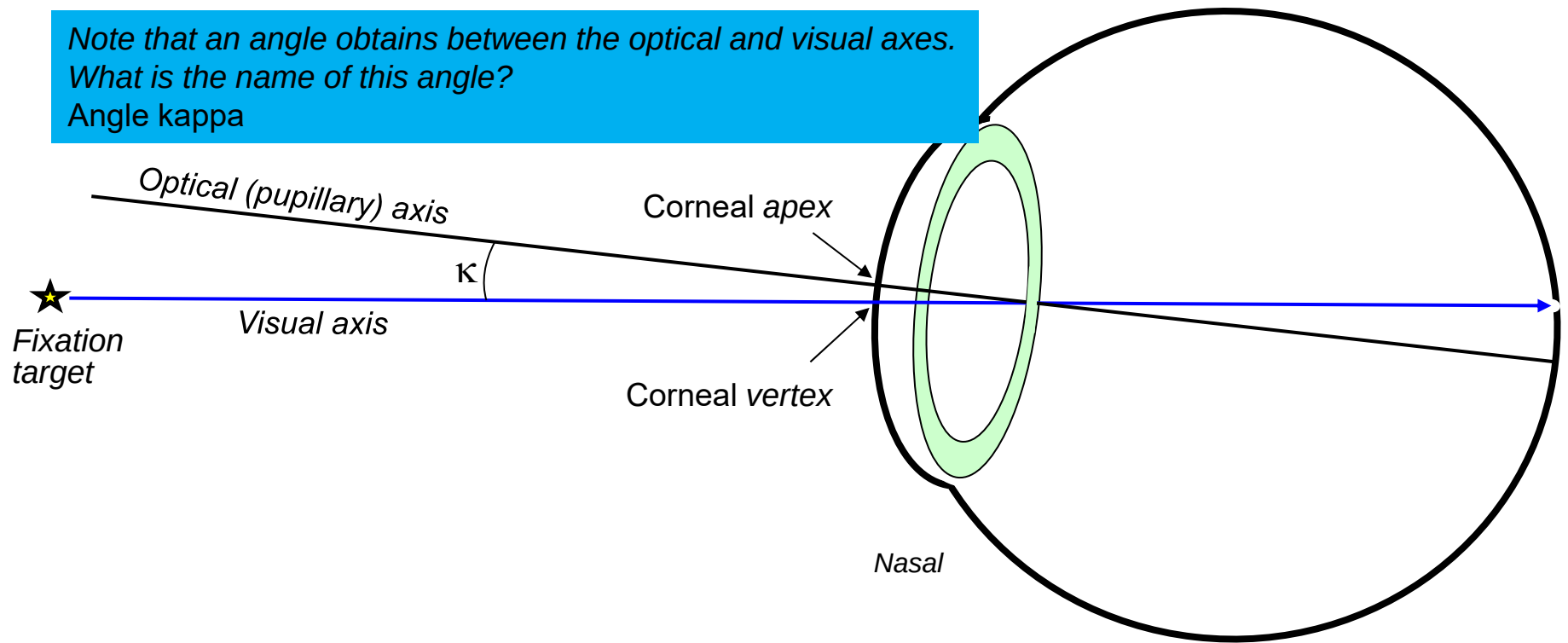
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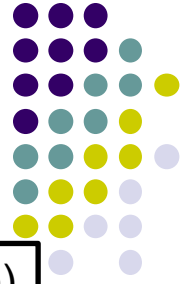
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Angle kappa



Q

Corneal Optics

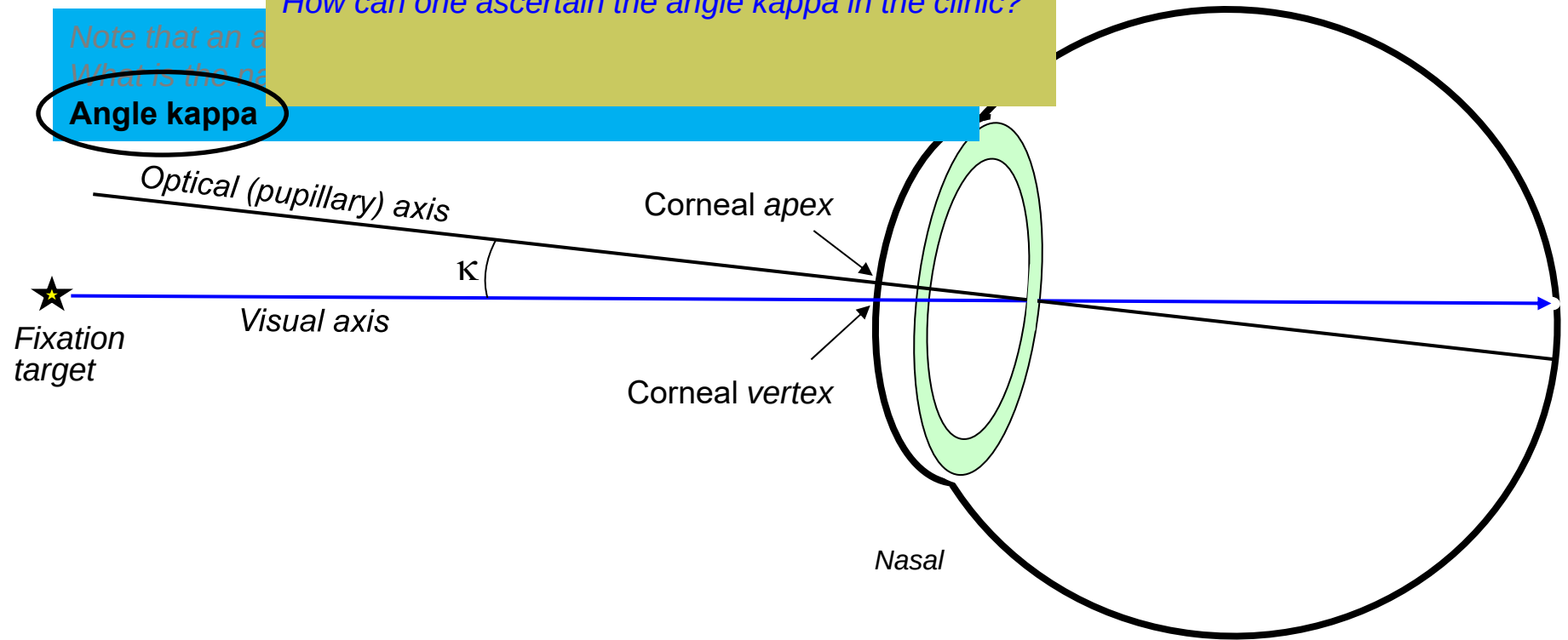


(OD, viewed from above)

How can one ascertain the angle kappa in the clinic?

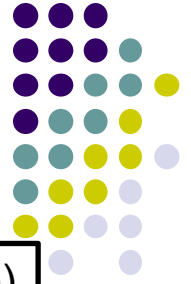
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A

Corneal Optics

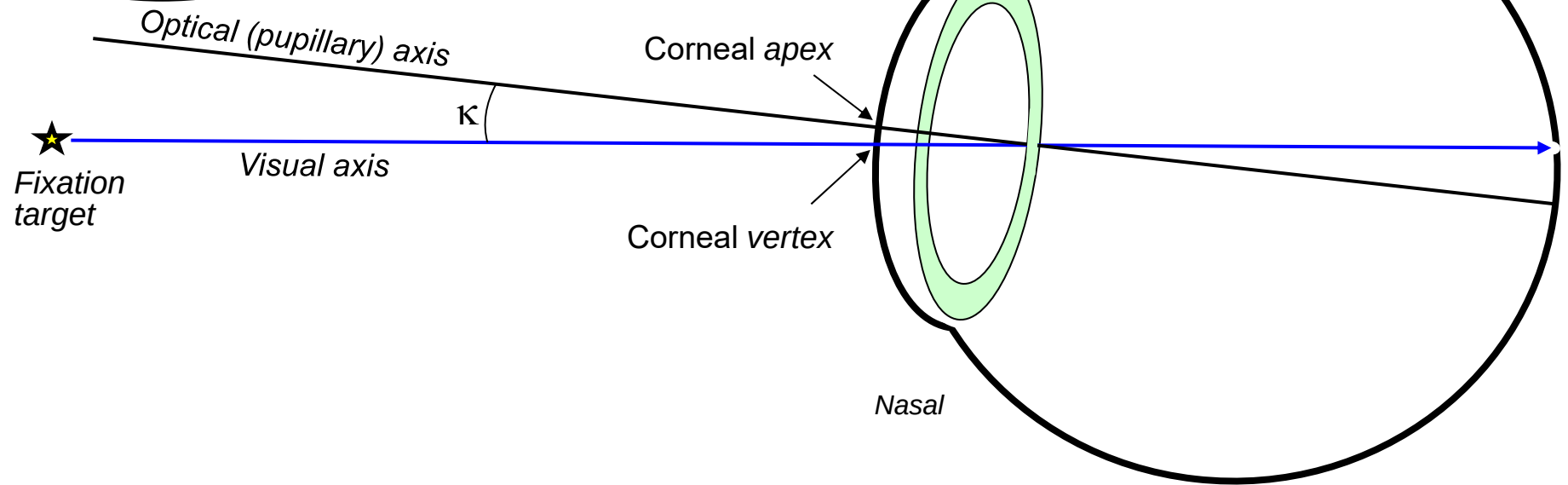


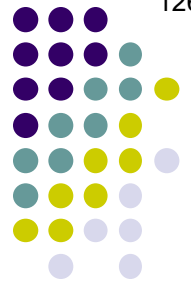
(OD, viewed from above)

How can one ascertain the angle kappa in the clinic?
By locating the corneal vertex, then taking note of its distance from the center of the pupil

Note that an angle kappa is present in all eyes.
What is the normal value of angle kappa?

Angle kappa





Q

Corneal Optics

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How does a corneal topographer work?

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Next Q

What sorts of artifacts interfere with interpretation of corneal topography?

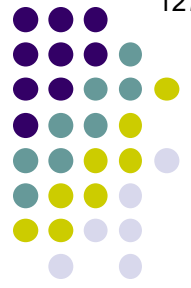
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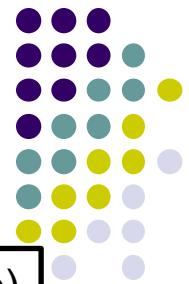
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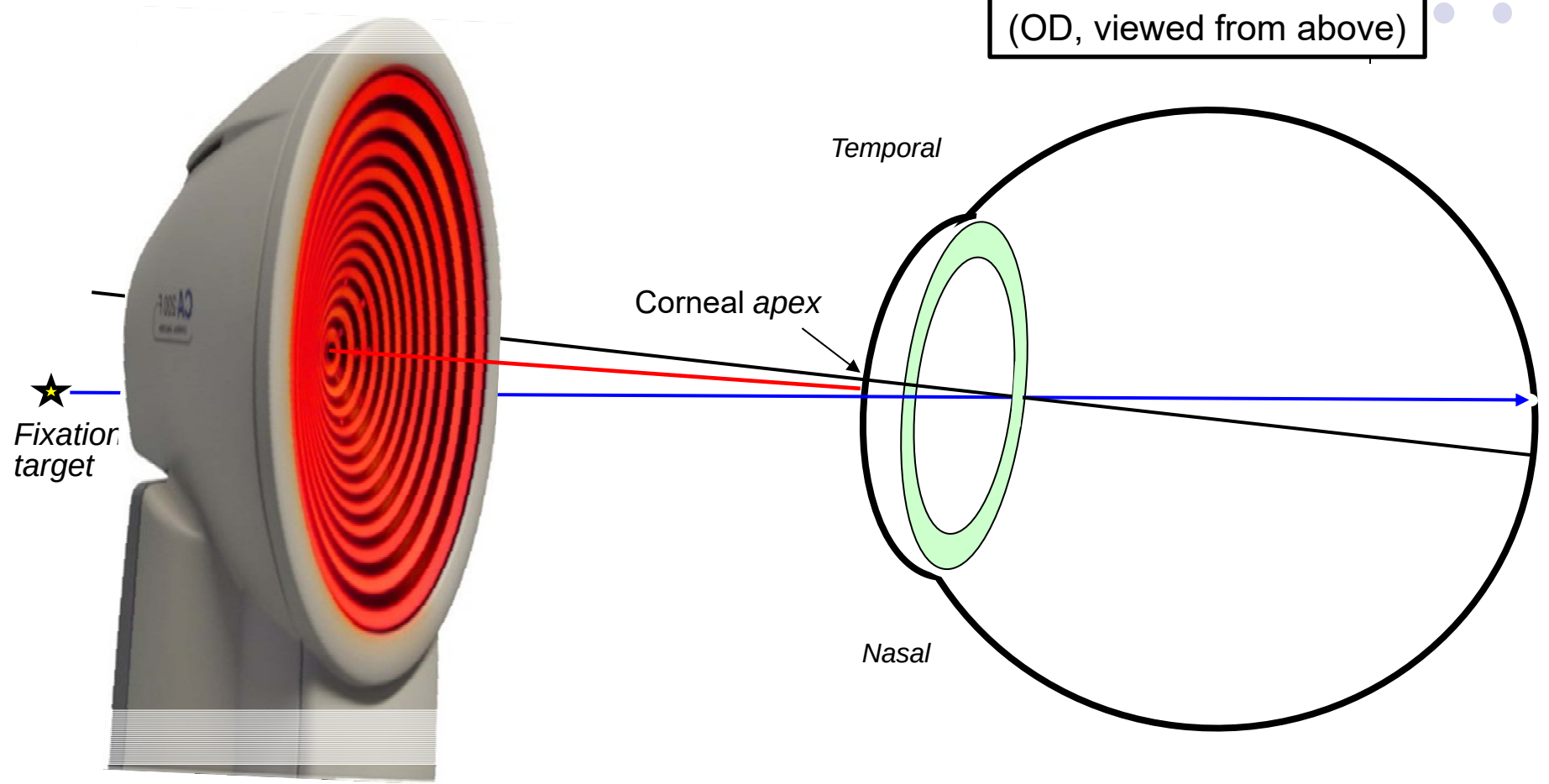
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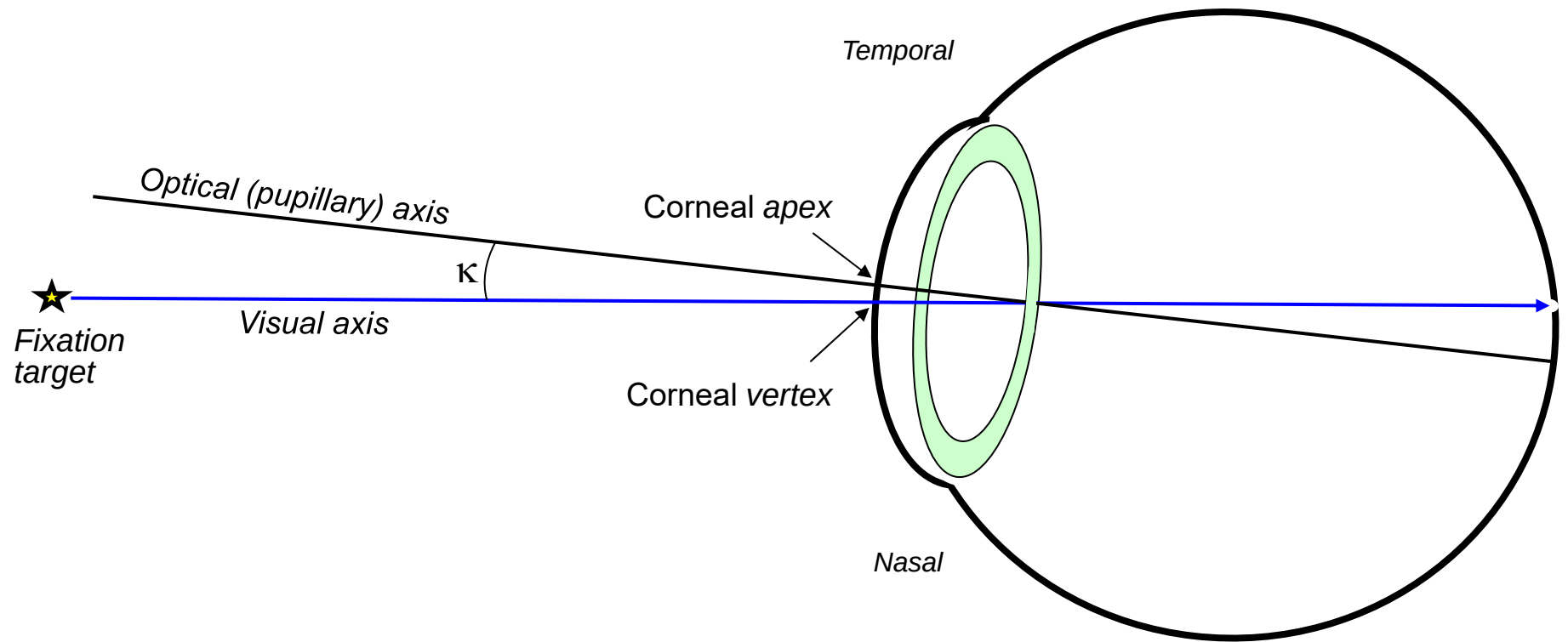


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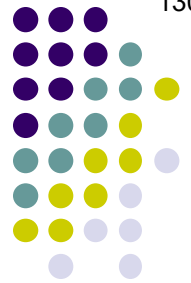
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Corneal Optics

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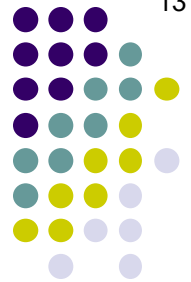
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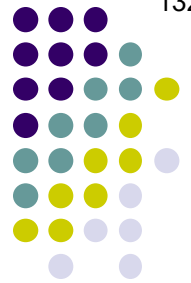
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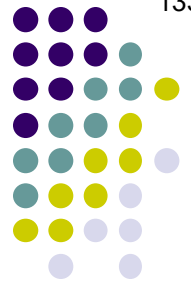
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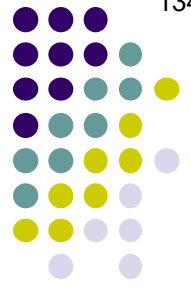
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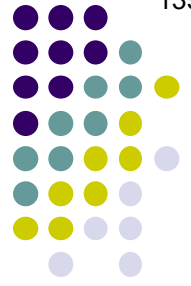
Corneal Optics



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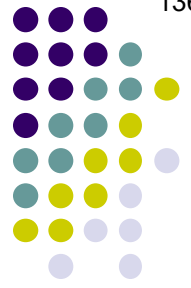
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Q

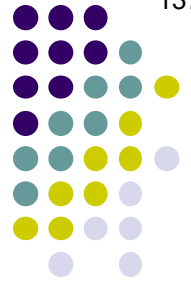
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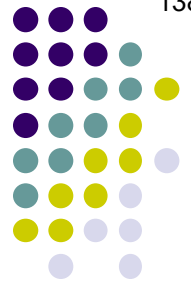
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Corneal Optics

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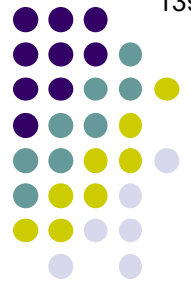
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There are two basic technologies:

- Scanning-slit
- Scheimpflug imaging



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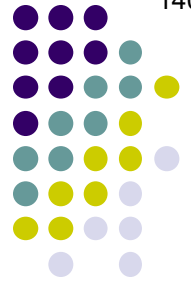
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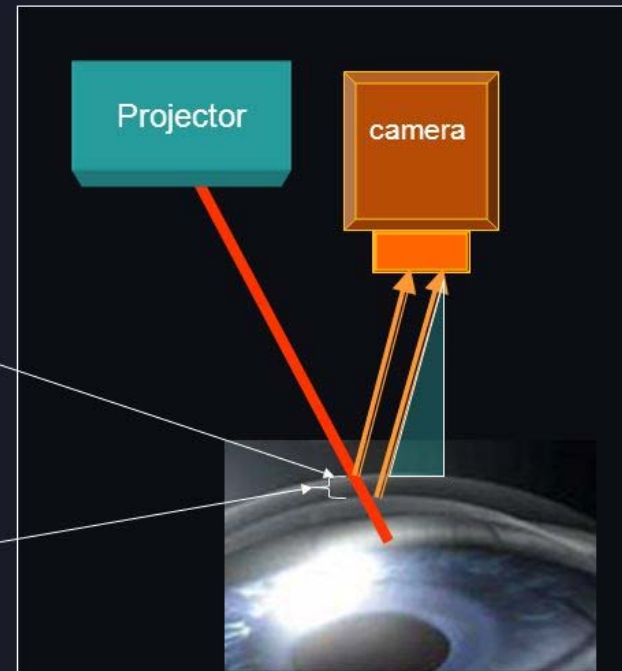
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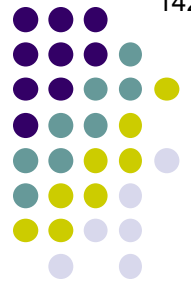
Multiple slit-beam images (which reflect off both the anterior *and* posterior corneal surfaces) are collected, and used to reconstruct the structure of the cornea. (Note: Some scanning-slit systems work by combining this technology with Placido-disk imaging)



Scanning slit technology

- The scanning slit triangulates the area between the reflected beam and camera axis to determine the edge.
- Overall analysis of reflected beams determines 'floats'.
- Difference in two edges created by the same beam, adjusting for time delays, determines the depth.





Q

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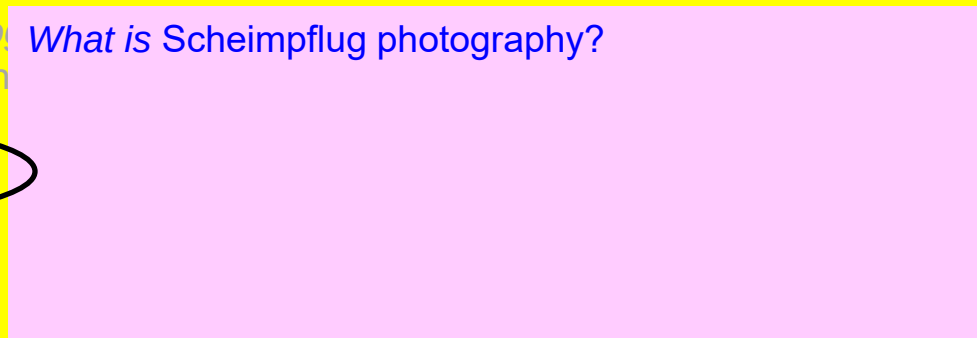
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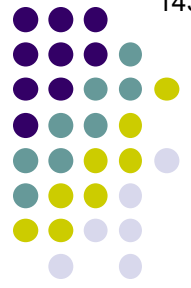
--**Scheimpflug imaging**

What is Scheimpflug photography?



A

Corneal Optics



- *What are the three categories of technology for determining central corneal power?*
 - Keratometry
 - Corneal topography
 - **Corneal tomography**

What is the difference between corneal topography and corneal tomography?

Unlike topography, which only provides information about the shape of the corneal *surface*, corneal tomography provides **3-D modeling** of the cornea--including anterior surface shape, posterior surface shape, and corneal thickness.

How does a corneal tomograph work?

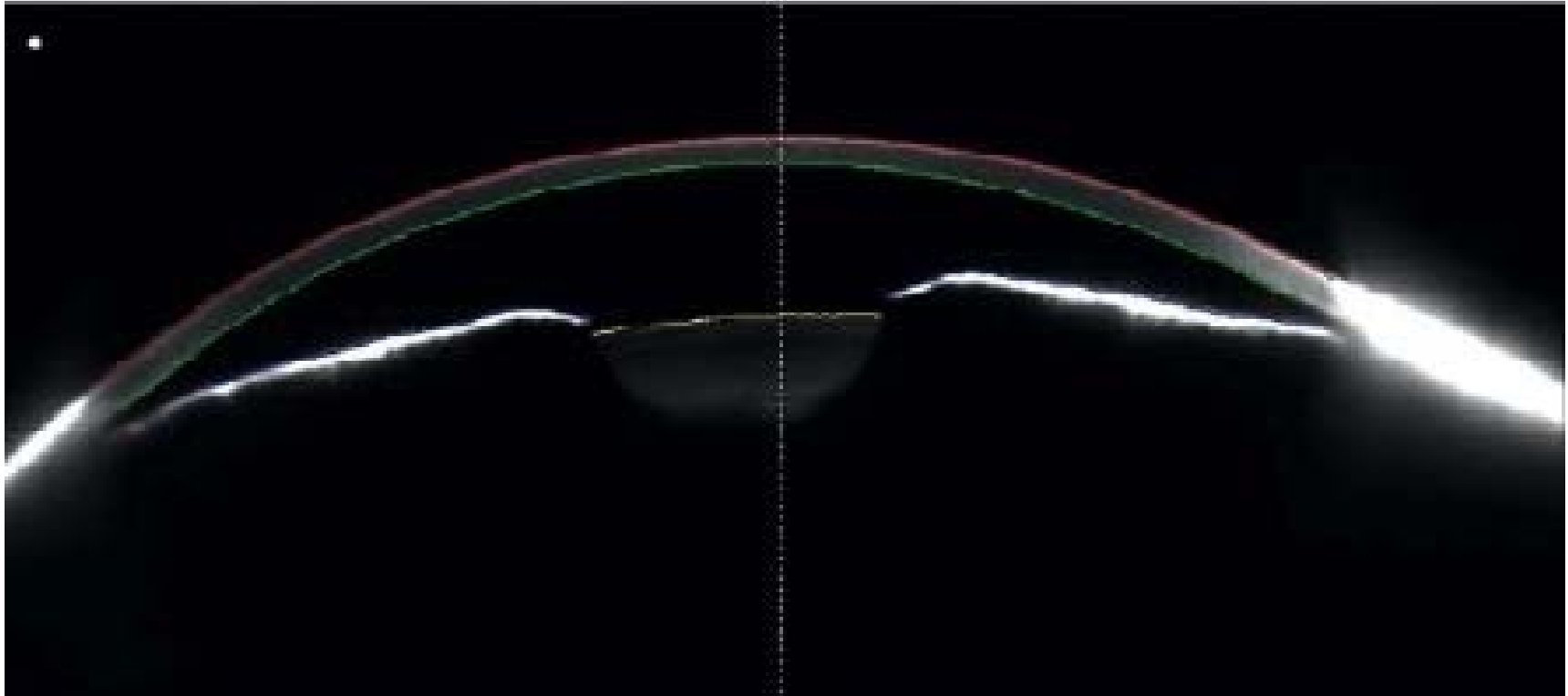
There are two basic techniques:

--Scanning slit

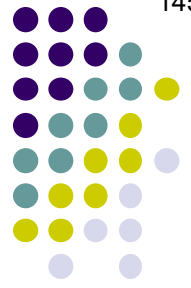
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A technique that allows one to take 'side on' pics of the cornea as slit-beam illumination passes through it



Corneal tomography: Scheimpflug image



Q

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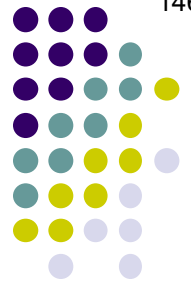
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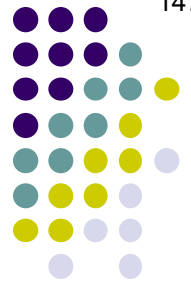
How do Scheimpflug-based systems work?

Multiple Scheimpflug images of the cornea are collected and integrated to produce a model of the cornea

Q

Corneal Optics

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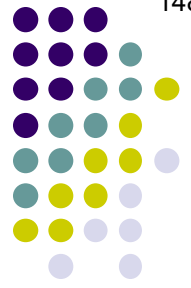
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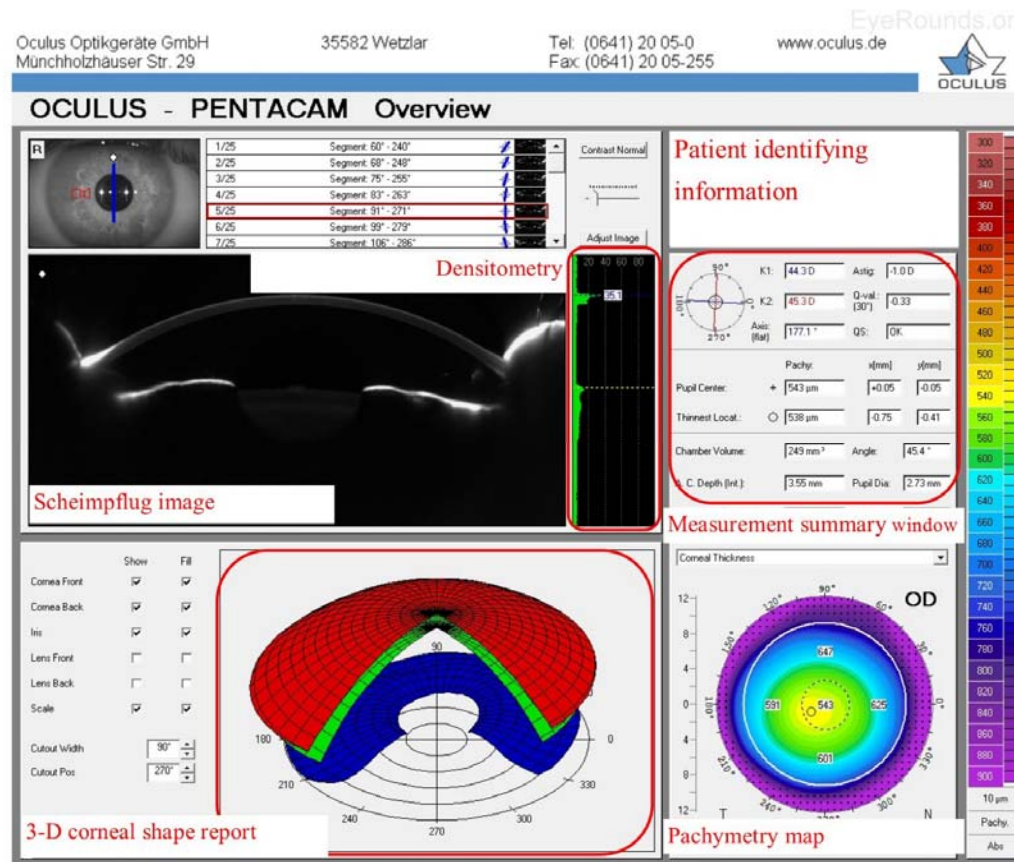
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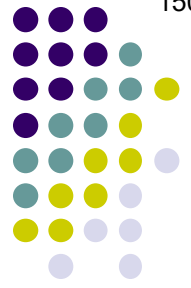
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Scheimpflug systems do; scanning-slit do not (but they estimate it very accurately)

Corneal Optics



Pentacam Overview Report. **Upper pane:** the Scheimpflug image is a cross-sectional image showing the cornea, anterior chamber, iris, and lens. **Lower panes:** A 3-D representation of the patient's corneal shape is provided; the anterior corneal surface is shown in red, posterior corneal surface in green, and iris in blue. **A pachymetry map is a color map that indicates corneal thickness; cooler colors are thicker and warmer colors are thinner (numeric scale on right).**



Q

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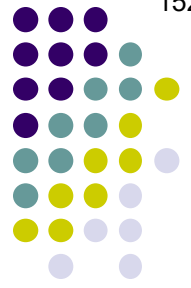
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Does tomography measure the curvature of the posterior corneal surface?

Yes (well, Scheimpflug systems do; again, scanning-slit systems estimate it)

Q

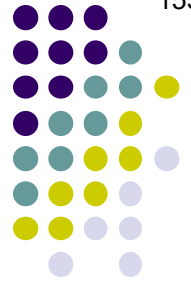
Corneal Optics



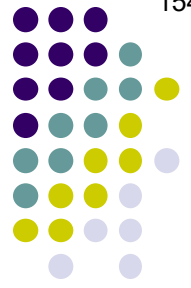
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Corneal Optics



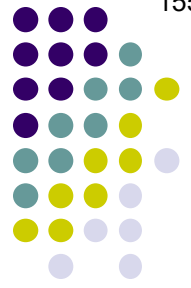
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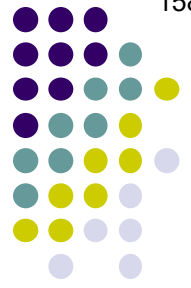
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 - Keratoconus
 - Pellucid marginal degeneration



Corneal Placido-disk topography: Mires typical of...



Corneal Placido-disk topography: Mires typical of...**keratoconus**



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- Corneal tomography

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For more on ectatic disorders, see slide-set K12

To determine whether a prospective pt suffers from a corneal ectatic disorder

- *What are the two main corneal ectatic disorders to be on the lookout for?*
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